

Rooted identity trees with fixed maximum outdegree

Boundary closure and computable asymptotic inverses

Research article 2 October 2026

Part II: 5 October 2026

Abstract

For each fixed integer $d \geq 2$, let $a_{d,n}$ count rooted unlabeled trees with at most d children at every vertex and trivial root-preserving automorphism group. We establish a unique square-root dominant singularity of their generating function by an explicit positive lower bound for the partial derivative in the signed root equation. A plane-tree majorant and a binary-root discriminant first give a finite boundary value and analytic nested inputs. The derivative bound then closes the characteristic-point and transversality arguments, while positivity after substitution excludes other dominant singularities. Consequently $a_{d,n} = C_d \rho_d^{-n} n^{-3/2} (1 + \sum_{j=1}^R D_{d,j} n^{-j} + \mathcal{O}_{d,R}(n^{-R-1}))$ for every fixed R .

We give finite Puiseux and Gamma-ratio generators, explicit first corrections for OEIS A116379 and A116380, and inverses of specified smooth models using W_{-1} or logarithmic polynomials. Integer thresholds receive an error enclosure, not unconditional exact rounding. The computational supplement contains exact enumeration and portable numerical replay. Numerical digits are stability checks, not interval certificates. Standard singularity methods and the derivative cancellation are prior work; the emphasis is the explicit analytic closure. Priority is not claimed, and two older full texts remain unchecked.

Part II (added 5 October 2026) prints a second, independent manuscript on the same theorem. Its new content is a rational-interval certificate, in exact integer arithmetic with Catalan tail bounds and a global identification of the critical root, which encloses ρ , τ , C , D_1 , D_2 , λ and the Puiseux coefficients $b_1, \dots, b_5$ for $d = 3, 4$ in intervals of width at most $1.4 \cdot 10^{-41}$. It certifies every digit of Table 1, so for these constants the digits are no longer only stability checks. Part II also proves $\rho_d < 0.4464105 < 9/20$ from an explicit infinite family, gives a purely rational recurrence for the Gamma-ratio coefficients and an exact threshold algorithm, and re-proves the all-orders theorem and the integer inverse enclosure by a second route. It corrects one citation of its source and adds a question on unrooted identity trees.

Results and conventions

- The cap is *at most* d children, including zero and one, rather than exactly d children at nonleaves. Distinctness is imposed among siblings at each vertex.
- The cases $d = 3$ and $d = 4$ are A116379 and A116380. On 2 October 2026 the inspected OEIS entries still describe the leading asymptotic as unknown.
- A degree-independent positive transversality bound is proved. All coefficient and inverse remainder statements nevertheless keep d and the expansion order fixed.
- The leading law, its corrections, and its inverses have separate logical roles. A smooth inverse is defined only after a specific finite model has been chosen.

[write] **Added 2 October 2026, batch 77.** This report is a single manuscript, printed in full: manuscript 63 of batch 77 of ProveIt’s incoming-reports intake, delivered as the archive `rooted-identity-trees-source.zip` (wrapper directory `rooted-identity-trees/`) in commit `096ee7b87`, and placed in commit `34f1acd4b` in the research-report collection, among the OEIS-sequence asymptotics reports (main file `report.tex`, now `article.tex`). The manuscript names no ProveIt commit and continues no repository path, so it has no pin; it names neither an author nor a tool, and its PDF metadata leave the author empty. Nothing in this report is formalized in Lean or Rocq, and placement in the collection confers no formal status on any statement here. No other report of the repository treats identity trees, A116379 or A116380. The writing step prefixed every label with `bit:` and added these [write] notes; no statement, proof or number of the manuscript was altered.

Priority caveat, kept. The manuscript makes no priority claim: the full texts of Meir–Moon–Mycielski (1983) and Haigh–Kennedy–Quintas (1984) were not inspected (Section 8), and the derivative cancellation is credited to Bell–Burris–Yeats. The statement that the OEIS entries “still describe the leading asymptotic as unknown” is the manuscript’s, dated 2 October 2026; the intake did not re-check the OEIS pages [re-checked 5 October 2026: both records still state that it is not known whether these trees have the asymptotic form $C\rho^{-n}n^{-3/2}$; see the end of Section 19].

Notation. The manuscript reuses several letters. Each symbol is printed as delivered; the table fixes its meaning by section, with the tempting false readings.

symbol	meaning here (and what it is <i>not</i>)
p, p_r	$p_r = I(z^r)$ are power sums (Sections 1–4); $p = 3/2$ is the polynomial exponent of the inverse model (35). The inverse sections never use p_r .
E_j, e_j, E_k	e_j elementary symmetric functions and $E_j = \sum_{k \leq j} e_k$ their cumulative sums (2); E_k in Proposition 6.1 are inverse coefficients, not cumulative sums.
P, P_k	$P(z)$ is the plane-tree (Catalan) majorant (8); $P_k(h)$ are the pure-log inverse polynomials (43)–(44).
B, B_j	$B = P(81/400)$ in the proof of Lemma 2.1 and a count-error constant in the proof of Proposition 7.1; $B_j(x)$ are Bernoulli polynomials.
h, h_r, H	$h = \log(L/\lambda)$ (Section 6); $h_r(\alpha)$ the log-Gamma coefficients (28); $H(z)$ the evaluated derivative (19).
d, D_j, Δ	d is the cap; $D_j = D_{d,j}$ the relative corrections ($D_0 = 1$); Δ the inverse displacement. D_j is not a derivative.
C, τ, ρ, λ	$C = C_d = b_1/\Gamma(-1/2)$; $\tau = I(\rho)$; $\lambda = \log(1/\rho)$. All depend on d ; Table 1 gives them for $d = 3, 4$.
$N, N_{r,q}, N(y)$	N is a nested-input truncation in the numerics; $N_{r,q}$ the nested coefficients (24); $N(y)$ the integer threshold (48).
$R, \mathcal{R}_R, K$	R is a model (expansion) order; $\mathcal{R}_R(n)$ the count residual (34); K an inverse order, $K_{d,R}$ the enclosure constant.
L, L_2, M_R, Q_R	$L = \log(y/C)$; $L_2 = D_2 - D_1^2/2$; M_R the smooth model and Q_R its correction factor.
t, η	$t = \sqrt{1 - z/\rho}$; $\eta = 1 - 810/(319\sqrt{19})$ the transversality margin.

[write] **Added 5 October 2026, batch 103.** Since 5 October 2026 this report has two Parts. Part I (Sections 1–10) is the batch-77 manuscript described in the preceding note, printed as before: its sections, equations, theorems, tables and labels keep their numbers. The preceding note’s “single manuscript, printed in full” and “no pin” now describe Part I; its statement that no other report treats identity trees, A116379 or A116380 remains true (the report `a182220-source-boundary`, written 3 October 2026, counts extensional acyclic digraphs, the other classical coding of hereditarily finite sets, and points to this one). Part II (Sections 11–19) prints bundle Report 130 of batch 103 (arrival commit `60f54ea06`, placed in `9c995cefe`), a second, independent manuscript of 2 October 2026 on the same theorem: unrefereed, presumed AI-assisted, not formalized, with no pin. Its new content is a rational-interval certificate of $\rho, \tau, C, D_1, D_2, \lambda$ and $b_1, \dots, b_5$ for $d = 3, 4$, which certifies the digits of Table 1. Part II’s symbols $F, G, E_j, P_j, T, B, H, \eta, L, u$ and M do not mean what the table above says; in particular Part II’s G is Part I’s F , and its F is minus Part I’s G . Section 11.3 fixes them.

Part I

Boundary closure and computable asymptotic inverses

1 The rooted set equation

A rooted identity tree is a finite rooted unlabeled tree whose only automorphism fixing the root is the identity. Its child subtrees are themselves identity trees and are pairwise nonisomorphic. Conversely, these two conditions force every root-preserving automorphism to be trivial: it cannot permute distinct child types and, recursively, cannot act inside a child. The restriction is siblingwise and recursive. The same fringe-tree type may occur at vertices with different parents.

Fix $d \geq 2$ throughout. Write

$$I(z) = I_d(z) = \sum_{n \geq 1} a_n z^n, \quad a_n = a_{d,n}, \quad a_0 = 0. \quad (1)$$

Size means number of vertices. There is a unary path at every positive size, so $a_n \geq 1$. Root deletion leaves a set of between zero and d distinct objects from the same class.

Define elementary symmetric polynomials in *power-sum coordinates* by

$$\sum_{j \geq 0} e_j(p_1, p_2, \dots) u^j = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1}}{r} p_r u^r \right), \quad E_j = \sum_{k=0}^j e_k. \quad (2)$$

Thus $e_0 = E_0 = 1$; put $E_j = 0$ for $j < 0$. When evaluating E_j below, its power sums are always stated or inherited from the displayed arguments. The root equation is

$$I(z) = F(z, I(z)), \quad F(z, y) = z E_d(y, I(z^2), \dots, I(z^d)). \quad (3)$$

Although the polynomial in (3) has signed coefficients, the series $e_j(I(z), I(z^2), \dots)$ counts sets of j distinct subtree objects and has nonnegative coefficients. Formally,

$$\sum_{j \geq 0} e_j(I(z), I(z^2), \dots) u^j = \prod_{n \geq 1} (1 + u z^n)^{a_n}. \quad (4)$$

This identity also gives an exact integer algorithm without cancellation.

For clarity, put $y = I(z)$ and $p_r = I(z^r)$. The two target equations are

$$d = 3: \quad y = z \left(1 + y + \frac{y^2 - p_2}{2} + \frac{y^3 - 3yp_2 + 2p_3}{6} \right), \quad (5)$$

$$d = 4: \quad y = z \left(1 + y + \frac{y^2 - p_2}{2} + \frac{y^3 - 3yp_2 + 2p_3}{6} + \frac{y^4 - 6y^2p_2 + 8yp_3 + 3p_2^2 - 6p_4}{24} \right). \quad (6)$$

These agree with A116379 and A116380 [7, 8]. For $d = 2$ the cubic and quartic terms are absent.

2 Boundary finiteness and a uniform derivative bound

Let ρ be the radius of I . We prove the analytic hypotheses directly, rather than assuming a positive-coefficient bivariate implicit schema for the signed polynomial F .

Lemma 2.1 (Radius and finite endpoint). *For every $d \geq 2$,*

$$\frac{1}{4} \leq \rho \leq \frac{9}{20}, \quad \tau := \lim_{z \uparrow \rho} I(z) = I(\rho) < \infty. \quad (7)$$

The functions $I(z^r)$, $r \geq 2$, needed in F are analytic on a neighborhood of the closed disk $|z| \leq \rho$.

Proof. Choose a canonical ordering of the children of each unlabeled tree, for example by recursively ordering their canonical encodings. This injects the identity-tree class into rooted plane trees. Hence coefficientwise

$$I(z) \preceq P(z) := \frac{1 - \sqrt{1 - 4z}}{2}, \quad P'(z) = \frac{1}{\sqrt{1 - 4z}}, \quad (8)$$

and $\rho \geq 1/4$.

Every class with cap $d \geq 2$ includes roots with zero children, one child, or two distinct children from the class. At $0 < z < \rho$, writing $y = I(z)$, this yields

$$y \geq z \left(1 + y + \frac{y^2 - I(z^2)}{2} \right). \quad (9)$$

Suppose $\rho > 9/20$. At $z = 9/20$ the necessary discriminant for this quadratic inequality would be nonnegative. Yet

$$\begin{aligned} (1 - z)^2 - 2z^2 + z^2 I(z^2) &\leq 1 - 2z - z^2 + z^2 P(z^2) \\ &= -\frac{10 + 81\sqrt{19}}{8000} < 0, \end{aligned} \quad (10)$$

a contradiction. This proves the non-strict upper bound without evaluating a boundary series.

Now let $z \uparrow \rho$. Since $z^2 \leq 81/400 < 1/4$, the term $I(z^2)$ is bounded by $B = P(81/400)$. On any interval $\rho/2 \leq z < \rho$, (9) implies $zy^2 \leq 2(1 - z)y + zB \leq 2y + zB$. The positive roots of this quadratic bound y uniformly because $z \geq 1/8$. Nonnegative coefficients therefore give a finite limit τ and convergence of $I(\rho)$ to that limit. Continuity would also exclude $\rho = 9/20$ by (10), but we retain the sufficient non-strict bound.

Finally $\rho^r \leq \rho^2 \leq 81/400 < 1/4 \leq \rho$ for $r \geq 2$. For fixed d there is a common disk larger than $|z| \leq \rho$ on which all finitely many nested inputs $I(z^r)$ are analytic. $\square$

[write] **Added 5 October 2026, batch 103.** Part II proves the strict bound explicitly (Section 13): an infinite injective family of binary identity trees, with generating function $z^{21}/(1 - z(1 + S(z)))$ where S collects the binary counts through size 20, gives $\rho_d \leq \rho_2 \leq r < 9/20$ for every $d \geq 2$, with $0.4464104 < r < 0.4464105$ ((66) and Remark 13.1). The non-strict bound proved here is all that Part I uses.

Differentiating (2) in p_r gives the useful cancellation

$$\frac{\partial E_j}{\partial p_r} = \frac{(-1)^{r-1}}{r} E_{j-r}. \quad (11)$$

For fixed-cardinality Set, Bell, Burris and Yeats [1, p. 57] already observe the $r = 1$ derivative cancellation and the resulting positivity after substitution. Equation (11) is the direct power-sum differentiation used here; neither it nor that positivity observation is claimed as new.

Proposition 2.2 (Explicit transversality). *On the physical positive branch and at its endpoint, the partial derivatives of F satisfy*

$$F_y(z, I(z)) = zE_{d-1}, \quad F_{yy}(z, I(z)) = zE_{d-2} \geq z, \quad (12)$$

$$F_z(z, I(z)) \geq \frac{I(z)}{z} \eta > 0, \quad \eta := 1 - \frac{810}{319\sqrt{19}} > 0. \quad (13)$$

Here $0 < z \leq \rho$, every E_j is evaluated at the actual subtree power sums, and F_z differentiates with respect to z while holding y fixed.

Proof. Equation (11) proves (12). Along the actual branch, (3) gives

$$F_z = \frac{y}{z} + \sum_{r=2}^d (-1)^{r-1} z^r I'(z^r) E_{d-r}. \quad (14)$$

All evaluated e_j are nonnegative. Thus $0 \leq E_{d-r} \leq E_d = y/z$, and (8) implies $I'(z^r) \leq P'(z^r)$. Discarding positive odd-index terms yields

$$\begin{aligned} F_z &\geq \frac{y}{z} \left(1 - \sum_{k \geq 1} z^{2k} P'(z^{2k}) \right) \\ &\geq \frac{y}{z} \left(1 - \frac{z^2}{(1-z^2)\sqrt{1-4z^2}} \right). \end{aligned} \quad (15)$$

The expression subtracted in the last line is increasing for $0 < z \leq 9/20$ and at $9/20$ equals $810/(319\sqrt{19}) < 1$. All nested derivatives are analytic at the endpoint, so the same inequality holds by continuity there. The estimate is independent of d and requires no prior characteristic equation. $\square$

3 The characteristic point and complex dominance

Theorem 3.1 (Unique square root singularity). *For each fixed $d \geq 2$, the point (ρ, τ) satisfies*

$$\tau = F(\rho, \tau), \quad F_y(\rho, \tau) = 1, \quad F_z(\rho, \tau) > 0, \quad F_{yy}(\rho, \tau) > 0. \quad (16)$$

It is the unique singularity of I on $|z| = \rho$. The function I continues analytically to a dented neighborhood of that disk and, locally at ρ , has the convergent expansion

$$I(z) = \sum_{j \geq 0} b_j (1 - z/\rho)^{j/2}, \quad b_0 = \tau, \quad b_1 = -\sqrt{\frac{2\rho F_z(\rho, \tau)}{F_{yy}(\rho, \tau)}} < 0, \quad (17)$$

Proof. For $0 < z < \rho$, differentiate (3):

$$(1 - F_y(z, I(z)))I'(z) = F_z(z, I(z)) > 0. \quad (18)$$

Since $I'(z) > 0$, we have $F_y < 1$. The finite endpoint and analytic nested inputs imply a finite endpoint limit with $F_y(\rho, \tau) \leq 1$. If it were less than one, the ordinary analytic implicit-function theorem applied to $G(z, y) = y - F(z, y)$ would continue I through the positive point ρ . This

contradicts Pringsheim's theorem for the nonnegative series I with finite radius. Thus $F_y = 1$. Proposition 2.2 proves the other inequalities in (16).

For complex dominance, define the *evaluated* derivative

$$H(z) = F_y(z, I(z)) = zE_{d-1}(I(z), I(z^2), \dots). \quad (19)$$

It has nonnegative coefficients. Its terms ze_0 and $ze_1 = zI(z)$ ensure positive coefficients of z and z^2 . Absolute convergence at ρ follows from $\tau < \infty$ and the finite polynomial expression; also $H(\rho) = 1$. If $z = \rho e^{i\theta}$, the triangle inequality gives $|H(z)| \leq 1$. Equality requires all nonzero terms to have the same argument. The terms of degrees one and two then require $e^{i\theta} = e^{2i\theta}$, so $e^{i\theta} = 1$. Consequently

$$|F_y(z, I(z))| < 1 \quad (|z| = \rho, z \neq \rho). \quad (20)$$

The series I converges absolutely and uniformly on its closed disk. At each such boundary point its value satisfies the analytic equation, and (20) allows implicit continuation across it.

At (ρ, τ) , $G_y = 0$, $G_z = -F_z \neq 0$ and $G_{yy} = -F_{yy} \neq 0$. Local analytic preparation therefore gives two square-root branches. Expanding G at that point gives $0 = \rho F_z t^2 - \frac{1}{2} F_{yy} (y - \tau)^2 + \mathcal{O}(t^3)$ with $t = \sqrt{1 - z/\rho}$. Because $I(z) < \tau$ for real $z < \rho$, the physical branch has the negative coefficient in (17). Nonzero b_1 makes ρ a genuine singularity.

Choose the local slit branch at ρ . The implicit continuations around the rest of the circle agree with the original interior function and with one another on their overlaps. Compactness of the circle away from a small neighborhood of ρ gives a common outer collar. Together these neighborhoods contain a usual Δ -domain:

$$|z| < \rho + \varepsilon, \quad z \neq \rho, \quad |\arg(z - \rho)| > \phi \quad (0 < \phi < \pi/2) \quad (21)$$

for sufficiently small $\varepsilon > 0$. This supplies the domain needed for singularity transfer [2, Chs. VI–VII]. $\square$

Two checks at the characteristic point are worth recording. Subtracting the root and derivative equations gives

$$\tau = 1 + \rho e_d > 1, \quad F_{yy} = \rho E_{d-2} = 1 - \rho e_{d-1} > 0. \quad (22)$$

The strict inequality $e_d > 0$ follows by choosing d distinct finite unary paths as child subtrees. These finite-cap identities are useful consistency tests; they do not constitute separate priority claims.

4 All fixed orders by finite coefficient operations

Theorem 4.1 (Coefficient expansion). *For every fixed $d \geq 2$ and every fixed integer $R \geq 0$,*

$$a_n = C \rho^{-n} n^{-3/2} \left(\sum_{j=0}^R D_j n^{-j} + \mathcal{O}_{d,R}(n^{-R-1}) \right), \quad D_0 = 1, \quad C = \frac{b_1}{\Gamma(-1/2)} > 0. \quad (23)$$

The constants b_j and D_j are determined by the finite recursions below. The domain, implicit constants and required lower bound on n may depend on d and R .

4.1 The local Puiseux generator

Put $t = \sqrt{1 - z/\rho}$, so $z = \rho(1 - t^2)$. For $r \geq 2$ define exact nested coefficients

$$N_{r,q} = (-1)^q \sum_{n \geq 1} a_n \rho^{rn} \binom{rn}{q}, \quad I(z^r) = \sum_{q \geq 0} N_{r,q} t^{2q}. \quad (24)$$

For each fixed q , the sum converges absolutely because $\rho^r < \rho$ and the binomial factor grows only polynomially in n . Equivalently, $N_{r,q}$ is the s^q coefficient of $I(\rho^r(1 - s)^r)$, giving a second numerical route by Taylor differentiation.

To compute through D_R , work modulo t^{2R+3} , retain $q \leq R + 1$ in (24), and use

$$e_0 = 1, \quad j e_j = \sum_{r=1}^j (-1)^{r-1} p_r e_{j-r}, \quad p_1 = y = \tau + \sum_{k \geq 1} b_k t^k. \quad (25)$$

For $r \geq 2$ substitute (24). After choosing the negative b_1 in (17), determine b_k successively for $2 \leq k \leq 2R + 1$ from

$$[t^{k+1}] \left(y - \rho(1 - t^2) \sum_{j=0}^d e_j \right) = 0. \quad (26)$$

At this stage the coefficient of the new unknown b_k is $-F_{yy} b_1 \neq 0$. Indeed the first relevant term of G_y along the expansion is $-F_{yy} b_1 t$, so varying $b_k t^k$ first changes degree $k + 1$ by that amount. A possible b_{k+1} term is multiplied by $G_y(\rho, \tau) = 0$. This proves that the recursion is triangular and uniquely solvable. Finite truncation of the sums in (24) is a numerical approximation to this exact definition.

4.2 The Gamma ratio generator

For noninteger α and $n \geq 0$,

$$[z^n](1 - z/\rho)^\alpha = \rho^{-n} \frac{\Gamma(n - \alpha)}{\Gamma(-\alpha)\Gamma(n + 1)}. \quad (27)$$

Let $B_j(x)$ denote the Bernoulli polynomials, with their usual exponential generating function $te^{xt}/(e^t - 1)$. Define $\gamma_0(\alpha) = 1$ and

$$h_r(\alpha) = \frac{(-1)^{r+1}}{r(r+1)} (B_{r+1}(-\alpha) - B_{r+1}(1)), \quad (28)$$

$$k\gamma_k(\alpha) = \sum_{r=1}^k r h_r(\alpha) \gamma_{k-r}(\alpha). \quad (29)$$

The log-Gamma expansion, followed by exponentiation, gives

$$\frac{\Gamma(n - \alpha)}{\Gamma(n + 1)} \sim n^{-\alpha-1} \sum_{k \geq 0} \gamma_k(\alpha) n^{-k}. \quad (30)$$

It follows that

$$D_j = \sum_{\ell=0}^j \frac{b_{2\ell+1}}{b_1} \frac{\Gamma(-1/2)}{\Gamma(-\ell - 1/2)} \gamma_{j-\ell}(\ell + 1/2). \quad (31)$$

In particular,

$$D_1 = \frac{3}{8} - \frac{3}{2} \frac{b_3}{b_1}, \quad (32)$$

$$D_2 = \frac{25}{128} - \frac{45}{16} \frac{b_3}{b_1} + \frac{15}{4} \frac{b_5}{b_1}. \quad (33)$$

[write] **Added 5 October 2026, batch 103.** Part II gives a second, purely rational recurrence (80) for these coefficients, without Bernoulli polynomials; Proposition 15.1 proves that it produces $\gamma_k(\alpha)$ (Part II's $H_k(\alpha)$) for every $\alpha \notin \{0, 1, 2, \dots\}$. Part II's formula (82) for its c_J is (31) term by term.

Proof of Theorem 4.1. The convergent Puiseux expansion from Theorem 3.1 may be truncated after t^{2R+2} with remainder $\mathcal{O}(t^{2R+3})$. The finitely many even powers are polynomials in z and have no coefficients for sufficiently large n . Transfer in the Δ -domain (21), applied to each odd power and to the remainder, yields an absolute remainder $\mathcal{O}_{d,R}(\rho^{-n}n^{-R-5/2})$. Substituting (30) into (27) and collecting powers of n^{-1} gives (31) and (23). This is the standard Puiseux and singularity-transfer mechanism; the preceding sections verify its hypotheses for the present signed equations. $\square$

Remark 4.2 (What the uniform bound does not prove). The number η in (13) is independent of d . This alone controls neither a common Δ -domain nor all local derivatives or transfer constants as $d \rightarrow \infty$. No asymptotic assertion here allows d or R to grow with n , and no convergence of the full inverse-power asymptotic series is asserted.

5 The ternary and quaternary cases

The characteristic system (16), with the nested functions evaluated from exact count polynomials, gives the numerical values in Table 1. Thirty decimal places are displayed to make comparisons unambiguous; none is interval-certified.

	$d = 3$ A116379	$d = 4$ A116380
ρ	0.400774500477539473704620732219	0.397585196670872770885924531230
τ	1.045166273395837405445129471289	1.006157697606778380774781545295
C	0.414316870666512014232537980552	0.371658807505686761191129677635
D_1	−0.708396139596070031595088414677	−0.453862309357400171863826908421
D_2	−0.572628680061486211786162046680	−0.562216000439483655162154638924

Table 1: Non-certified numerical characteristic data and first relative corrections.

For example, writing the displayed decimals only as approximations,

$$a_{3,n} \approx 0.4143168707 (0.4007745005)^{-n} n^{-3/2} \left(1 - \frac{0.7083961396}{n} - \frac{0.5726286801}{n^2} \right),$$

$$a_{4,n} \approx 0.3716588075 (0.3975851967)^{-n} n^{-3/2} \left(1 - \frac{0.4538623094}{n} - \frac{0.5622160004}{n^2} \right).$$

Numerical work should use the stored high-precision constants, since a rounded base raised to the n th power eventually introduces a material error.

[write] **Added 2 October 2026, batch 77.** The stored values in `data/identity-checks.json` carry 48 decimals. The table rounds them correctly, so in two entries its last digit differs from the 30-decimal prefix of

the file: C for $d = 3$ is stored as $\dots 980551907\dots$ (printed $\dots 980552$), and ρ for $d = 4$ as $\dots 531229552\dots$ (printed $\dots 531230$). The intake checked the table independently from the exact counts: at $n = 400$, $n(a_n/(C\rho^{-n}n^{-3/2}) - 1) - D_1$ is -0.00146 for $d = 3$ and -0.00143 for $d = 4$, matching $D_2/n = -0.00143$ and -0.00141 to the expected accuracy, which confirms ρ , C , D_1 and D_2 for both caps.

[write] **Added 5 October 2026, batch 103.** Part II (Section 16) certifies these constants. Its rational-interval certificate (exact integer arithmetic with outward rounding, Catalan tail bounds for the nested jets, and a global argument that identifies the critical root on $0 < z \leq 9/20$) encloses ρ , τ , C , D_1 , D_2 , and also λ , $b_1, \dots, b_5$, F_z and F_{yy} , for $d = 3, 4$, in intervals of width at most $1.4 \cdot 10^{-41}$ (Table 6). Every entry of Table 1 lies strictly within half a unit of its last decimal from the corresponding enclosure, so each printed digit is the correctly rounded value; the same holds for the E_1 , E_2 values of Section 6 (Remark 16.2). The caption “Non-certified” and the statements that no value is interval-certified describe the manuscript as delivered and are kept. The certificate is computer-assisted (a Python program in exact integer arithmetic), not a proof-assistant check, and it uses Theorem 3.1 to identify the endpoint. D_j for $j \geq 3$ and the residual Table 2 are not certified.

Define the residual, including its sign convention, by

$$\mathcal{R}_R(n) := \frac{a_n}{C\rho^{-n}n^{-3/2} \sum_{j=0}^R D_j n^{-j}} - 1. \quad (34)$$

A negative residual means the approximation is larger than the exact count. Table 2 uses full stored precision.

Cap	n	$\mathcal{R}_0(n)$	$\mathcal{R}_2(n)$	$\mathcal{R}_4(n)$
3	100	$-7.146115 \cdot 10^{-3}$	$-4.925663 \cdot 10^{-6}$	$-3.441364 \cdot 10^{-8}$
3	200	$-3.556885 \cdot 10^{-3}$	$-5.903010 \cdot 10^{-7}$	$-1.002226 \cdot 10^{-9}$
3	400	$-1.774641 \cdot 10^{-3}$	$-7.231917 \cdot 10^{-8}$	$-3.030477 \cdot 10^{-11}$
4	100	$-4.598866 \cdot 10^{-3}$	$-4.040312 \cdot 10^{-6}$	$-3.061665 \cdot 10^{-8}$
4	200	$-2.283849 \cdot 10^{-3}$	$-4.836327 \cdot 10^{-7}$	$-8.922404 \cdot 10^{-10}$
4	400	$-1.138229 \cdot 10^{-3}$	$-5.921972 \cdot 10^{-8}$	$-2.698812 \cdot 10^{-11}$

Table 2: Exact count divided by truncated asymptotic model, minus one.

The factor close to eight on doubling n in the order-two column is consistent with an n^{-3} relative remainder. This observation is a diagnostic, not the proof of Theorem 4.1.

Exact counts were computed through $n = 400$. Nested jets were evaluated by direct weighted binomial sums with $(N, \text{digits}) = (200, 70)$ and $(400, 110)$, and by Taylor differentiation of the composed polynomial with $(N, \text{digits}) = (250, 80)$. The resulting $\rho, \tau, C, D_0, \dots, D_4$ agree to more than 60 decimal places. These are independent constructions of the *nested jets*; they share the exact count data, characteristic-point procedure and later jet algebra. They are not independent full proofs or rigorous enclosures of truncation and rounding errors.

6 Inverting a specified smooth asymptotic model

A sequence on the integers has no canonical smooth inverse. We therefore choose a model before inverting it. Fix a finite $R \geq 0$, set

$$p = \frac{3}{2}, \quad \lambda = \log(1/\rho) > 0, \quad Q_R(x) = 1 + \sum_{j=1}^R D_j x^{-j}, \quad M_R(x) = C e^{\lambda x} x^{-p} Q_R(x). \quad (35)$$

For this model only, set $D_j = 0$ for all $j > R$ whenever a higher inverse order is requested. This is a convention about the chosen model, not a statement that the true sequence corrections vanish.

For sufficiently large real x , $Q_R(x) > 0$ and

$$(\log M_R)'(x) = \lambda - \frac{p}{x} + \mathcal{O}(x^{-2}) > \lambda/2. \quad (36)$$

Thus the restriction of M_R to a sufficiently large half-line has a unique inverse $x_R(y)$ for all sufficiently large y .

6.1 The exact Lambert carrier

The $R = 0$ model has large inverse

$$x_0(y) = -\frac{p}{\lambda} W_{-1} \left(-\frac{\lambda}{p} (C/y)^{1/p} \right). \quad (37)$$

Indeed taking the p th root of $y = Ce^{\lambda x} x^{-p}$ gives $(-\lambda x/p)e^{-\lambda x/p} = -(\lambda/p)(C/y)^{1/p}$. The argument lies in $(-1/e, 0)$ when y is large. The lower real branch W_{-1} gives the solution tending to infinity; W_0 gives the small solution. Formula (37) is exact for M_0 on its increasing branch.

Proposition 6.1 (Recursive Lambert corrections). *For the specified finite model (35) and each fixed $K \geq 0$,*

$$x_R(y) = x_0(y) + \sum_{k=1}^K E_k x_0(y)^{-k} + \mathcal{O}_{d,R,K}(x_0(y)^{-K-1}). \quad (38)$$

The coefficients are generated by setting $u = 1/x_0$, $\Delta = \sum_{k \geq 1} E_k u^k$ and cancelling powers of u in

$$\lambda \Delta - p \log(1 + u\Delta) + \log \left(1 + \sum_{j=1}^R D_j \left(\frac{u}{1 + u\Delta} \right)^j \right) = 0. \quad (39)$$

At degree k , the only occurrence of the new coefficient is λE_k .

Proof. Subtract $\log M_0(x_0) = \log y$ from $\log M_R(x_0 + \Delta) = \log y$ to obtain (39). Its left side is analytic near $(u, \Delta) = (0, 0)$ and has partial derivative λ in Δ there. The implicit-function theorem gives an analytic solution $\Delta(u)$ with zero constant term, proving both the finite-order remainder and the triangular recurrence. $\square$

Writing $L_2 = D_2 - D_1^2/2$ and using the zero-extension convention if $R < 2$, the first coefficients are

$$E_1 = -\frac{D_1}{\lambda}, \quad E_2 = -\frac{pD_1}{\lambda^2} - \frac{L_2}{\lambda}. \quad (40)$$

For any model of order at least two, these are approximately

	E_1	E_2
$d = 3$	0.77474841994368976130	2.1716520579717961183
$d = 4$	0.49207378938552388788	1.5214703882822106514

When $K \leq R$, $E_1, \dots, E_K$ depend only on the true $D_1, \dots, D_K$ and agree with formal inversion of the full asymptotic expansion. When comparing a sequence to its order- R model, $K = R$ keeps model and inverse truncation errors at the same order.

6.2 A pure logarithmic expansion

Let

$$L = \log(y/C), \quad h = \log(L/\lambda). \quad (41)$$

For each fixed K , the same large inverse can be written without evaluating Lambert W :

$$\lambda x_R(y) = L + ph + \sum_{k=1}^K \frac{P_k(h)}{L^k} + \mathcal{O}_{d,R,K} \left(\frac{(\log L)^{K+1}}{L^{K+1}} \right). \quad (42)$$

Here P_k is a polynomial of degree at most k . The first two are

$$P_1(h) = p^2 h - \lambda D_1, \quad (43)$$

$$P_2(h) = -\frac{p^3}{2} h^2 + p^3 h + p\lambda D_1(h-1) - \lambda^2(D_2 - D_1^2/2). \quad (44)$$

To generate higher P_k , substitute $X = L + ph + \sum P_k(h)L^{-k}$ into

$$X - p \log(X/\lambda) + \log \left(1 + \sum_{j=1}^R D_j (\lambda/X)^j \right) = L. \quad (45)$$

At each new order the unknown P_k has coefficient one. Treat h as a formal indeterminate during cancellation, then put $h = \log(L/\lambda)$. Expanding $\log(1 + \mathcal{O}(\log L/L))$ shows inductively that $\deg P_k \leq k$. The residual after order K has the order in (42). The derivative of the left side of (45) with respect to X tends to one, so the solution error has the same order. This gives a remainder statement, rather than only a formal series.

6.3 Numerical separation of inverse errors

For the order-four model let $\hat{x}_4 = x_0 + \sum_{k=1}^4 E_k x_0^{-k}$. At the exact targets $y = a_n$, record three quantities with their indicated signs:

$$x_4(a_n) - n, \quad \hat{x}_4(a_n) - x_4(a_n), \quad \hat{x}_4(a_n) - n. \quad (46)$$

They separate the sequence-to-model discrepancy, the inverse-series discrepancy, and their sum. The supplement records them for $n = 50, 100, 200, 400$. At $n = 400$, the last quantity is approximately $-4.1616 \cdot 10^{-11}$ for $d = 3$ and $-3.6307 \cdot 10^{-11}$ for $d = 4$. These tests concern real inverse location, not a certified ceiling at a discontinuity.

7 Integer thresholds and the ceiling qualification

Adding a new unary root gives an injection from the size- n class to the size- $(n+1)$ class and preserves the cap and identity property. Hence $a_{n+1} \geq a_n$. The leading theorem gives

$$\frac{a_{n+1}}{a_n} \longrightarrow \rho^{-1} > 1, \quad (47)$$

so the sequence is eventually strictly increasing. For sufficiently large real y , define

$$N(y) = \min\{n \in \mathbb{N} : a_n \geq y\}. \quad (48)$$

Proposition 7.1 (Qualified integer enclosure). *For fixed d and R , there is a constant $K_{d,R} > 0$ such that, for all sufficiently large y , writing $x = x_R(y)$,*

$$\left\lceil x - K_{d,R}x^{-R-1} \right\rceil \leq N(y) \leq \left\lceil x + K_{d,R}x^{-R-1} \right\rceil. \quad (49)$$

The same form holds with x replaced by the order- R Lambert-corrected approximation, after increasing the constant.

Proof. Theorem 4.1 and positivity of Q_R for large arguments give $|\log a_n - \log M_R(n)| \leq Bn^{-R-1}$ for large integer n . Equation (36) bounds the logarithmic model slope below by $\lambda/2$. Choose $K_{d,R}$ sufficiently larger than $2B/\lambda$ and put $\delta = K_{d,R}x^{-R-1}$. For integers near x , if $n < x - \delta$, the model slope dominates the count error and $a_n < y$; if $n \geq x + \delta$, then $a_n \geq y$. Integers far below x are handled by eventual monotonicity, with the finite initial segment smaller than y for large y . Thus no integer smaller than $\lceil x - \delta \rceil$ can attain the threshold, while $\lceil x + \delta \rceil$ does. This proves (49). Proposition 6.1 with $K = R$ changes x by at most a constant multiple of x^{-R-1} , which can be absorbed by enlarging $K_{d,R}$. $\square$

An exact ceiling is justified only when both enclosing ceilings in (49) coincide. A finite asymptotic expansion cannot decide every threshold arbitrarily close to a jump. In particular, at $y = a_n$ even a tiny negative inverse error must not be interpreted as an unconditional rounding theorem. No effective numerical value of $K_{d,R}$ or certified finite starting threshold is claimed here.

[write] **Added 2 October 2026, batch 77.** The inversion of Sections 6–7 is an instance of general results of the repository’s transseries volume (Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/), and no novelty is claimed for the method. The Lambert carrier (37) is `p0:thm:lambert-core` for $\lambda x - p \log x = \log(y/C)$, that is $a = \lambda$, $b = -p = -3/2$, with the branch W_{-1} that its branch rule selects for the large solution. The recursion (39) is the fixed-point equation of `p0:thm:lambert-centered` with $a = \lambda$, $b = -p$, $t = u = 1/x_0$, $U = \Delta$ and $Q(t) = \log(1 + \sum_{j \leq R} D_j t^j)$, which also gives the triangular structure noted after Proposition 6.1. The pure logarithmic expansion (42) is of the kind treated by `p0:thm:flattening` (direct polynomial–logarithmic reversion); the intake did not match the two normalizations term by term. The enclosure (49) has the form of the separation condition, part (2) of `p0:thm:staircase`: equal ceilings determine $N(y)$, as the paragraph above says. The generic staircase arithmetic is formalized in Lean as `Fabius.staircase_ceil` and `Fabius.staircase_separation` (Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean); those lemmas concern an arbitrary monotone function, not $a_{d,n}$, and give this report no formal status.

[write] **Added 5 October 2026, batch 103.** Part II’s Theorem 17.1 is the same integer enclosure in another normalization ($\log y$ and $\log \log y$ instead of $L = \log(y/C)$ and h), with the pure-logarithmic approximant directly inside the ceilings; Remark 17.2 derives it from Proposition 7.1 and (42), and its polynomials (99)–(100) are (43)–(44) after re-expansion. Part II’s exact threshold algorithm (Section 18) decides $N(y)$ for a given y by integer comparisons alone. Neither Part gives an effective $K_{d,R}$ or a certified starting threshold.

8 Relation to prior work and remaining priority checks

The leading $n^{-3/2}$ mechanism, analytic implicit-function analysis, Puiseux expansions and Gamma-ratio transfer are classical [2]. Bell–Burris–Yeats [1, §§6.1–6.2] explain why the signed Set or PSET construction is not automatically covered by their broad positive-operator theorem. Their Example 77 treats binary identity trees separately, and Example 78 discusses the unrestricted class. They already state the derivative cancellation for fixed-cardinality Set and the resulting

nonnegativity after substitution on printed page 57. The issue addressed by Sections 2–3 is the remaining finite-boundary, nonvanishing-derivative and complex-dominance closure, with an explicit bound valid for every finite cap.

Genitrini’s all-order Pólya-structure analysis [3] assumes an equation $T = \zeta \exp T$ with suitable analyticity beyond the dominant radius and a nonzero derivative of ζ . Its identity-tree example in §3.2 is the unrestricted PSET class A004111, not either finite cap considered here. Defining $\zeta = I \exp(-I)$ formally would not by itself verify those analytic hypotheses and cannot replace the boundary argument. The degree-restricted Ω -Pólya framework in Gittenberger–Jin–Wallner [4, §5.2] is based on multiset trees and explicitly distinguishes identity trees; it does not supply a positive-weight bounded-PSET theorem.

The inspected A116379 and A116380 pages retain an unknown-asymptotic comment as of 2 October 2026 [7, 8]. That database status motivates the calculation but is not evidence of exhaustive priority. Two especially relevant older works remain unresolved at full-text level: Meir–Moon–Mycielski’s *Hereditarily finite sets and identity trees* [5], and Haigh–Kennedy–Quintas’s *Counting and coding identity trees with fixed diameter and bounded degree* [6]. The latter’s available abstract concerns large-diameter asymptotics, but this does not exclude an additional size theorem in the complete paper. Public publisher and metadata routes did not yield readable full texts in the present check. Accordingly this article makes no first-ever claim or assertion that those papers omit the result. Establishing publication-level novelty would require resolving those sources and a broader literature review.

[write] **Added 5 October 2026, batch 103.** Part II’s manuscript cites two further older works whose full texts it did not obtain, Labelle (1992) [9] and Kennedy–McKeon–Palmer–Robinson (1990) [10], and attributes [6] to “G. W. Furnas”, which is wrong (Remark 11.1); it does not cite Meir–Moon–Mycielski. The priority question, now with four unchecked papers, is question 5 of Section 19.

9 Exact enumeration and reproducibility

The accompanying archive is self-contained apart from standard software dependencies. It contains this article’s TeX and PDF, source code, generated numerical data, build instructions and a file-hash manifest. It contains no third-party full papers.

For exact enumeration, suppose $a_1, \dots, a_{n-1}$ are known. The number a_n is the sum, over $0 \leq j \leq d$, of the coefficient of $u^j z^{n-1}$ in

$$\prod_{m=1}^{n-1} (1 + uz^m)^{a_m}. \quad (50)$$

Truncate the u degree at d and the z degree at the requested size. Multiplying the factor for size m uses the exact binomial coefficients $\binom{a_m}{\ell}$ and descending updates in u degree. Since a child has strictly fewer vertices than its parent, the construction determines all counts successively from $a_1 = 1$. A separately implemented Newton recurrence from (25) provides an exact cross-check.

The numerical stage solves the two characteristic equations for the finite nested-input polynomial, forms the jets by (24) or Taylor differentiation, and applies (26) and (31). The inverse stage implements (39) and (45), solves the large model inverse directly, and separates the three residuals in (46). The code supports finite orders beyond the displayed examples; their numerical reliability still depends on precision and nested-input truncation.

The replay script is invoked with `bash replay.sh`. It regenerates exact counts, cross-checks the two enumeration routes, reproduces the jet and inverse tests, and rebuilds the article. The README

specifies the software versions and commands, expected outputs, residual conventions and the limits of numerical verification. A SHA-256 manifest identifies the released inputs and outputs. On a different TeX engine or package version PDF bytes may differ; in that case mathematical content and rendering should be compared rather than equating a different metadata or font encoding with changed results.

[write] **Added 2 October 2026, batch 77.** This section describes the delivered archive. In the repository `code/` and `data/` keep their delivered names, while `replay.sh` and `build_pdf.sh` moved into `code/`, `code/requirements.txt` into `data/`, and `report.tex` became `article.tex`. The delivered PDF and the SHA-256 manifest `SHA256SUMS` were verified at placement and not shipped (both survive in the arrival commit). `replay.sh` begins by checking that manifest and then rebuilds `report.pdf`, so it cannot run as shipped; the numerical replay `code/reproduce.sh` needs neither and runs with the shipped layout (the report's `README.md` gives the command, with an output directory outside the repository). At intake (2 October 2026) it passed every check in 41 s, and `check_identity.py` reproduced the stored counts.

10 Further research questions

1. **Certified constants and effective thresholds.** Certify ρ, C, D_j by interval root and tail bounds, then make (49) effective.

[write] **Added 5 October 2026, batch 103.** The first half is answered for $d = 3, 4$ and $j \leq 2$: Part II (Section 16, Table 6) certifies ρ, C, D_1, D_2 (and $\tau, \lambda, b_1, \dots, b_5$) by exactly such an interval root and tail-bound computation. Open: D_j for $j \geq 3$ and caps other than 3 and 4 (the method is specified there, not implemented), and the second half, an effective (49) with explicit $K_{d,R}$ and starting threshold. See questions 1–2 of Section 19.

2. **Growing degree caps.** Determine whether the local analytic domains and derivative estimates can be controlled uniformly as $d \rightarrow \infty$, and identify the scale at which finite-cap coefficients approach unrestricted identity-tree coefficients. The bound in Proposition 2.2 is only one ingredient.
3. **Other allowed outdegree sets.** For arbitrary finite allowed sets the cumulative E_d structure is lost. Which restrictions retain enough positivity of the evaluated derivatives for a comparable closure? Periodicity and absent unary roots would require separate dominance and threshold arguments.
4. **Structure of random bounded identity trees.** Derive height, profile or fringe-subtree laws from suitable marked equations after verifying their signed analytic hypotheses. Existing multiset-tree limit laws cannot simply be imported.
5. **Priority and formulation.** Resolve the two older full-text references, compare maximum graph degree with rooted outdegree conventions, and identify the precise strongest statement already in the literature before making a novelty claim.

[write] **Added 5 October 2026, batch 103.** Question 2 is unchanged by Part II and is its question 3. Question 5 now concerns four older papers (question 5 of Section 19). Questions 3 and 4 are untouched. Part II adds a question on unrooted identity trees with a bound on the total degree (its question 4).

Part II

Certified constants and a second route

Part I (Sections 1–10) is unchanged except for dated [write] notes of 5 October 2026 (after the first note, after the proof of Lemma 2.1, after (33), in Section 5, at the end of Section 7, in Section 8, and in and after the list of Section 10), a paragraph of the abstract, a line in the title block, the heading of Part I, and two bibliography entries added for Part II; after the independent check of the write (5 October 2026), the first note and the entries [5], [7] and [8] carry dated additions. No number of Part I moves.

11 Part II: source, scope and notation

This section was written when Part II was added. Sections 12–18 print the manuscript; Section 19 collects its unproved claims and open directions together with those that remain of Part I. An independent check of the write, made afterwards, is recorded at the end of Section 19.

11.1 Source and status

Sections 12–18 print bundle Report 130 of ProveIt’s incoming-reports intake (batch 103), titled “All-orders asymptotics and inversion for bounded-outdegree rooted identity trees” and dated 2 October 2026, the same day as the manuscript of Part I. It arrived as the archive A116379_A116380_All_Orders_Identity_Trees_and_Inversion_Source.zip (main file Report130.tex, which inputs certified_constants.tex; 17-page PDF) in commit 60f54ea06 and was placed in commit 9c995cefe, its code and data under the local prefix 02-certified-. Its author line, and the author field of its PDF, read “Research report”; like Part I it names neither an author nor a tool and is presumed AI-assisted. It calls itself standalone, names no ProveIt commit and does not know Part I, so it has no pin. It is a second, independent answer to the same question: its main theorem is the theorem of Part I. Part II is unrefereed; nothing in it is formalized in Lean or Rocq, and its place in the collection confers no formal status. The manuscript’s Sections 1–10 and Appendices A–B are Sections 12–18 here: its Sections 1–2 form Section 12, Section 3 is Section 13, Sections 4–5 are Section 14, Sections 6–7 and Appendices A–B are Section 15, Section 8 with the input file certified_constants.tex is Section 16, Section 9 is Section 17, and Section 10 is Section 18 except its “Further research” paragraph, which opens Section 19. Every statement, proof, table and number of the manuscript is printed; wording was shortened where a passage repeats Part I verbatim in substance, and such places say so.

New in Part II (relative to Part I): the rational-interval certificate of Section 16, which encloses $\rho, \tau, C, c_1, c_2, \lambda, b$ and $\beta_1, \dots, \beta_5$ for $d = 3, 4$ and identifies the critical root globally on $0 < z \leq 9/20$; the strict bound $\rho_d < 9/20$ from an explicit infinite family (Section 13); the purely rational recurrence (80) for the Gamma-ratio coefficients; the explicit β_2, β_3 and the t^2, t^3, t^4 equations; the exact threshold algorithm of Section 18; and the question on unrooted identity trees (Section 19). *Second routes* to Part I: Theorem 12.1 is Lemma 2.1 with Theorems 3.1 and 4.1; c_J is D_J of (31); and Theorem 17.1 is a corollary of Proposition 7.1 with (42) (Remark 17.2). Table 3 maps every result.

The manuscript’s limitations are printed with it and remain in force: it makes no literature-priority

claim (its title footnote: “No publication or priority claim is made”; its Section 1: the agreement with the OEIS records is “*not a certification of literature priority*”); the derivative identity is credited to Bell–Burris–Yeats and the unrestricted all-orders expansion to Genitrini, neither claimed as new; the full texts of Labelle (1992) and Kennedy–McKeon–Palmer–Robinson (1990) were not obtained, so incidental overlap is not excluded; no estimate is uniform as $d \rightarrow \infty$; the certificate uses the qualitative theorem to identify the radius and does not independently prove analytic continuation; it is computer-assisted, not a proof-assistant development; its executable is restricted to $d = 3, 4$, $N = 400$ and local degree six, and the arbitrary-order extension is specified, not implemented; the 90-place strings of its JSON record the arithmetic grid, not 90 certified digits; no finite- n error bound, explicit inverse remainder constant or certified starting threshold is given, and $N_d(T) \neq f_M(\log T) + o(1)$ in general; hashes give file identity, not authenticity; and nothing was uploaded, submitted or published.

11.2 What was rechecked at intake and in this write

Independently of the delivery, on 5 October 2026 (Python 3.14.4; the release used 3.12.14):

- at intake, the 14 entries of the delivered checksum manifest and the closed-inventory check of `verify_package.py` passed; its `-replay` fails on Windows for a line-ending reason only (the programs print in text mode, so Windows writes CRLF where the reference files have LF), and every replayed output equals its reference after removing carriage returns; the negative regression suite `test_package.py` passed in ordinary and optimized mode;
- in this write, the certifier and its supplementary tests (`code/02-certified-certify.py`, `code/02-certified-test_certifier.py`), copied to a scratch directory with the prefix removed, reproduced `data/02-certified-certificate.json` and `data/02-certified-tests.json` up to carriage returns; the coefficient exports for $d = 3, 4$ have the SHA-256 recorded in the certificate (after LF normalization) and equal the counts $a_{d,0}, \dots, a_{d,400}$ of Part I’s `data/identity-checks.json`;
- the certifier was read in full; the global bounds (89), the constants $27/80$, $787/2000$, $787/4000$, the value $p(9/20) = 7879/40000$, the table of $a_2(n)$ and the rational number (65) were recomputed exactly;
- (80) was compared with Part I’s Bernoulli route (28)–(29) symbolically in γ for $m \leq 5$ and exactly for seven rational γ and $m \leq 10$ (all equal; Proposition 15.1 proves equality in general), and the polynomials (99)–(100) were compared with Part I’s (43)–(44) after re-expansion (SymPy 1.14.0; equal);
- every entry of Part I’s Table 1 and of its E_1, E_2 table was compared with the certified enclosures by exact rational arithmetic (Remark 16.2);
- the bibliographic data of the paper the manuscript credits to “G. W. Furnas” were checked against Crossref (Remark 11.1).

11.3 Reading conventions

Part II prints the manuscript’s symbols unchanged. Several of them clash with Part I; the table fixes them. The most dangerous clash is the *swap of F and G* : Part II’s G is Part I’s F , and Part II’s $F = G - y$ is *minus* Part I’s $G = y - F$. Partial derivatives agree up to that renaming (G_z, G_{yy} here are F_z, F_{yy} there).

Part II symbol	meaning in Part II, and the Part I counterpart (or the false reading)
$A_d, a_d(n)$	generating function and counts; Part I's $I_d, a_{d,n}$.
$G(z, y), F(z, y)$	$G = z \sum_{j \leq d} P_j$ is Part I's F ; $F = G - y$ is $-G$ of Part I. Not Part I's F .
E_j, Φ_j	$E_j(z)$: series of sets of j distinct trees, Part I's e_j evaluated at the tree power sums; $\Phi_j = \sum_{i \leq j} E_i$ is Part I's cumulative E_j . Part II's E_j is <i>not</i> Part I's E_j , nor Part I's inverse coefficients E_k .
$P_j(z, y), P_m(\ell)$	elementary polynomials in y with inner $A(z^k)$ (56); and the inverse polynomials (98), in $\ell = \log \log T$ (not Part I's $P_k(h)$, $h = \log(L/\lambda)$, nor the plane-tree $P(z)$).
$T(z), T_{k,j}, T$	plane-tree (Catalan) series, Part I's $P(z)$; inner jets (87), with $T_{k,j} = (-1)^j N_{k,j}$ for Part I's $N_{r,q}$ (24); and the threshold target T of Section 17.
β_m, c_j, d_j	Puiseux coefficients, Part I's b_m ; relative corrections, Part I's D_j ; d_j the logarithmic corrections (96) (d_2 is Part I's L_2), not the cap d .
C, ρ, τ, λ	as in Part I; $\lambda = \log(1/\rho)$.
b, α, u	$b = -\alpha \log \lambda - \log C$ and $\alpha = 3/2$ (Part I's p); $u = \alpha \ell + b$ in Section 17, but $u = z^2 A'(z^2)$ in Section 16. Not Part I's Puiseux b_j .
$B(x), B_k$	$B(x) = x\Phi_{d-1}(x)$, Part I's H (19); $B_k(z) = A(z^k)$, Part I's p_k . Not Bernoulli polynomials or Part I's constant B .
$H(z), H_m(\gamma)$	the spine subclass (64); Gamma-ratio coefficients (79), equal to Part I's $\gamma_m(\gamma)$ (Proposition 15.1). Not Part I's $H(z)$.
η, η_j, ϕ	radial enlargement of the Δ -domain (Part I's ε); tail ratios (88). Not Part I's transversality margin η .
L, ℓ	$L = \log T$, $\ell = \log L$; Part I's L is $\log(y/C)$.
$N, N_d(T), M, K, R$	$N = 400$ the enumeration bound; $N_d(T)$ the threshold, Part I's $N(y)$; M the inverse order, the doubling bound of the threshold algorithm, and $M(z)$ of (92); K the bracket constant (Part I's $K_{d,R}$); R an interval containing ρ , and R_{m+1} the residuals of (76) (not Part I's model order R).
Q, Z, J	$Q(y)$ a local analytic factor; $Z(y)$ the local inverse of A ; J the expansion order (Part I's R).

Bibliography keys were merged with Part I's: the manuscript's BBY, **Genitrini**, **FS**, **oeis3**, **oeis4** are Part I's [1], [3], [2], [7], [8], and its **Furnas** is Part I's [6] (Remark 11.1). Its two further references are [9] and [10].

Part II	Part I	status
Theorem 12.1 (59)	Lemma 2.1, Theorems 3.1, 4.1 (11)	same theorem, second proof same identity (BBY, p. 57)
(64)–(66)	Lemma 2.1 (non-strict)	second route; strict, explicit
(71)	Proposition 2.2	weaker bound, at the critical point only
Section 14.4 (76)	proof of Theorem 3.1 (26)	same argument same recurrence; (77)–(78) new displays
(80)	(28)–(29)	new formula, same numbers
(82)–(84)	(31)–(33)	same formulas
(85)–(86)	(5)–(6)	same equations
Section 16, Tables 4–6	Table 1 (non-certified)	new: certified; Part I's digits inside
Theorem 17.1, (101)	Proposition 7.1, (42), (45)	same theorem, other normalization
exact threshold algorithm	Section 9 (enumeration only)	new, elementary

Table 3: Map of Part II to Part I.

Remark 11.1 (A wrong citation, corrected). The manuscript's bibliography lists *Counting and coding identity trees with fixed diameter and bounded degree*, Discrete Applied Mathematics **7** (1984), 141–160, doi:10.1016/0166-218X(84)90063-5, as a paper by “G. W. Furnas”. Its authors are C. W. Haigh, J. W. Kennedy and L. V. Quintas: so says the Crossref record of that DOI (checked 5 October 2026), and Part I already cites the paper correctly as [6]. The DOI, title, journal, volume and pages

printed by the manuscript are right; only the author is wrong. Part II cites the paper as [6]. Its full text has not been inspected for either Part.

12 The main theorem and the signed set equation

A *rooted identity tree* is an isomorphism class of finite rooted trees whose root-preserving automorphism group is trivial. Children are unordered and vertices are unlabelled. Fix an integer $d \geq 2$, and require that every vertex have at most d children. The bound is on *outdegree*; it is not a bound of d on the total degree of a nonroot vertex. Let $A_d(z) = \sum_{n \geq 1} a_d(n)z^n$. All limits keep d fixed; no uniform estimate as $d \rightarrow \infty$ is asserted.

The model has a direct recursive interpretation: an object consists of a root and a set of at most d pairwise nonisomorphic child trees of the same class. Equivalently it is a hereditarily finite set with cardinality at most d at every nested set, size counted by the total number of recursively written set nodes. This is a natural regular family of combinatorial sequences, rather than a sequence built to manufacture a particular asymptotic law.

Theorem 12.1 (Complete asymptotic expansion). *For each fixed $d \geq 2$, there exist numbers $\rho = \rho_d$, $\tau = \tau_d$, $C = C_d$ and real constants $c_j = c_{d,j}$ such that*

$$\frac{1}{4} \leq \rho < \frac{9}{20}, \quad \tau = A_d(\rho) < \infty, \quad C > 0.$$

The point ρ is the unique singularity on the circle of convergence. There are $\eta > 0$ and $0 < \phi < \pi/2$ such that A_d continues analytically to

$$\Delta = \{z : |z| < \rho + \eta, \ z \neq \rho, \ |\arg(z - \rho)| > \phi\}. \quad (51)$$

At ρ , it has a convergent expansion

$$A_d(z) = \tau + \sum_{m \geq 1} \beta_m (1 - z/\rho)^{m/2}, \quad \beta_1 < 0. \quad (52)$$

For every fixed integer $J \geq 0$,

$$a_d(n) = C\rho^{-n}n^{-3/2} \left(1 + \sum_{j=1}^J c_j n^{-j} + O_{d,J}(n^{-J-1}) \right). \quad (53)$$

The constants are specified by convergent inner series and finite recurrences in Sections 14.4–15.

Apart from the strict inequality $\rho < 9/20$, this is Lemma 2.1 with Theorems 3.1 and 4.1 of Part I ($\beta_m = b_m$, $c_j = D_j$, $J = R$); the proof below is a second route to it.

The cases $d = 3$ and $d = 4$ are A116379 and A116380. The manuscript states that the records inspected on 2 October 2026 explicitly retain a question about the leading form $C\rho^{-n}n^{-3/2}$ [7, 8], which Theorem 12.1 answers in the rooted, maximum-outdegree model; this concerns the contents of those records and the proof, and is *not a certification of literature priority*. (Part I reports the same OEIS status. The intake re-checked both records on 5 October 2026: they still state that it is not known whether these trees have the asymptotic form $C\rho^{-n}n^{-3/2}$ [updated at the independent check; see the end of Section 19].) Bell, Burris and Yeats already developed the signed-SET derivative identities and treated binary identity trees in their discussion of mixed-sign functional equations

[1, Section 6]; Genitrini gives full expansions for the *unrestricted* identity-tree class [3]. Neither is claimed as new. Related work includes bounded-degree, fixed-diameter identity trees [6], asymmetric enriched trees [9] and symmetries in locally restricted trees [10]; the last two full texts were not obtained, so incidental overlap has not been excluded. The proof does not depend on an absence claim about that literature.

12.1 The exact signed set equation

An automorphism can act within a child subtree or permute isomorphic child subtrees, so a rooted tree is an identity tree if and only if all its child subtrees are identity trees and no two are isomorphic. Suppress the subscript d and put

$$E_j(z) = [u^j] \prod_{n \geq 1} (1 + uz^n)^{a_d(n)}, \quad E_0(z) = 1, \quad \Phi_j(z) = \sum_{i=0}^j E_i(z). \quad (54)$$

The coefficient of $u^j z^m$ counts sets of j distinct allowed trees with m vertices in total; all coefficients of E_j and Φ_j are nonnegative integers. The root decomposition gives

$$A(z) = z\Phi_d(z), \quad (55)$$

which determines the coefficients successively. The logarithm of the product in (54) is

$$\sum_{k \geq 1} \frac{(-1)^{k-1}}{k} u^k A(z^k).$$

With an independent outer variable y , set

$$P_j(z, y) = [u^j] \exp\left(uy + \sum_{k=2}^d \frac{(-1)^{k-1}}{k} u^k A(z^k)\right), \quad (56)$$

$$G(z, y) = z \sum_{j=0}^d P_j(z, y), \quad F(z, y) = G(z, y) - y. \quad (57)$$

Terms with $k > d$ cannot affect these coefficients. G is a polynomial in y whose coefficients involve the inner functions $A(z^k)$; it need not have nonnegative coefficients as a bivariate series. Nevertheless

$$F(z, A(z)) = 0. \quad (58)$$

Differentiation in the independent y shifts the index of u , so for $0 \leq r \leq d$

$$\partial_y^r G(z, A(z)) = z\Phi_{d-r}(z); \quad (59)$$

in particular $G_y(z, A(z)) = z\Phi_{d-1}(z)$ and $G_{yy}(z, A(z)) = z\Phi_{d-2}(z)$. Positivity is claimed *after substituting the actual tree series*, not for arbitrary values of the outer variable; this distinction lets the proof work with the signed SET equation. (This is (11) of Part I with $r = 1$ iterated; both credit Bell–Burris–Yeats.) Newton’s identities give an exact coefficient algorithm:

$$jE_j(z) = \sum_{k=1}^j (-1)^{k-1} A(z^k) E_{j-k}(z), \quad j \geq 1. \quad (60)$$

At degree m the right side involves only $a_d(n)$ with $n \leq m$, so computing the forest coefficients and then $a_d(m+1) = [z^m]\Phi_d(z)$ is a noncircular recurrence (Part I’s (25)).

13 A strict radius bound from an explicit infinite family

Choosing a canonical ordering of the children of every rooted nonplane tree (for example by recursively ordering canonical parenthesis encodings) injects the class into rooted plane trees, so

$$1 \leq a_d(n) \leq \text{Cat}(n-1) \leq 4^{n-1}, \quad (61)$$

the lower bound being the rooted path. Plane rooted trees have generating function

$$T(z) = \sum_{n \geq 1} \text{Cat}(n-1)z^n = \frac{1 - \sqrt{1-4z}}{2}, \quad (62)$$

so $\rho_d \geq 1/4$ (Part I's (8)).

For a strict upper bound valid for every $d \geq 2$, a finite count is promoted to an *infinite injective subclass*. The binary identity-tree equation $A_2 = z(1 + A_2 + \frac{1}{2}(A_2(z)^2 - A_2(z^2)))$ gives $a_2(1) = 1$ and, for $n \geq 2$,

$$a_2(n) = a_2(n-1) + \frac{1}{2} \left(\sum_{i=1}^{n-2} a_2(i)a_2(n-1-i) - \mathbf{1}_{2|n-1} a_2\left(\frac{n-1}{2}\right) \right), \quad (63)$$

an empty sum being zero. It yields the complete finite data

n	1	2	3	4	5	6	7	8	9	10
$a_2(n)$	1	1	1	2	3	6	11	22	43	88
n	11	12	13	14	15	16	17	18	19	20
$a_2(n)$	179	372	774	1631	3448	7347	15713	33791	72923	158021

(equal to the $d = 2$ counts of Part I's `data/identity-checks.json`). Let $S(z) = \sum_{n=1}^{20} a_2(n)z^n$. Start from one fixed rooted path with 21 vertices. At each step add a new root, with either the current tree as its only child or with two children: the current tree and any binary identity tree of size at most 20. The children in the second case are distinct because their sizes differ. This encoding is injective: at every added root there is a unique child of size at least 21, and at most one other child, of size at most 20. Decode by removing roots and following the unique large child until the fixed 21-vertex seed is reached; a nonempty sequence of extensions has more than 21 vertices, so it cannot be confused with the seed. The subclass therefore has generating function

$$H(z) = \frac{z^{21}}{1 - z(1 + S(z))}, \quad H(z) \preceq A_2(z) \preceq A_d(z), \quad (64)$$

where $\preceq$ is coefficientwise domination. Exact rational evaluation gives

$$\frac{9}{20} \left(1 + S\left(\frac{9}{20}\right) \right) - 1 = \frac{64127583713420512656095449}{2097152000000000000000000000} > 0. \quad (65)$$

The function $x(1 + S(x))$ increases strictly from zero on $x \geq 0$, so its unique positive solution of $x(1 + S(x)) = 1$ is some $r < 9/20$; the denominator of H vanishes there without cancellation, so H has a singularity at r , and coefficientwise domination proves

$$\frac{1}{4} \leq \rho_d \leq \rho_2 \leq r < \frac{9}{20}. \quad (66)$$

This is a radius proof from an infinite family, not an extrapolation from a finite list of coefficient ratios.

Remark 13.1 ([write] The value of r). Exact rational evaluation of $x(1+S(x))-1$ at $x = 0.4464104$ and 0.4464105 gives a negative and a positive number, so $0.4464104 < r < 0.4464105$ and $\rho_d < 0.4464105$ for every $d \geq 2$. Part I's uncertified $\rho_2 = 0.42892377\dots$ (`data/identity-checks.json`) is consistent with this. Part I's Lemma 2.1 proves $\rho \leq 9/20$ by the discriminant (10) and notes that continuity excludes equality; (66) is a second, explicit route with a strictly smaller bound. Part II needs the strict bound only through $\rho^2 < 1/4$, which $\rho \leq 9/20$ already gives.

14 The critical endpoint, transversality and dominance

14.1 Finiteness and the characteristic equations

Because forests of exactly two trees are among those of at most d , for $0 < x < \rho$ we have $A(x) \geq \frac{x}{2}(A(x)^2 - A(x^2))$, that is $xA(x)^2 - 2A(x) - xA(x^2) \leq 0$, and therefore

$$A(x) \leq \frac{1 + \sqrt{1 + x^2 A(x^2)}}{x}. \quad (67)$$

Since $\rho < 1$, $A(\rho^2)$ is finite, and on $\rho/2 \leq x < \rho$ the right side of (67) is uniformly bounded. Monotone convergence gives $\tau := \lim_{x \uparrow \rho} A(x) = \sum_{n \geq 1} a_d(n) \rho^n < \infty$. Every inner function $A(z^k)$, $k \geq 2$, is analytic on $|z| < \sqrt{\rho}$, a disk strictly larger than $|z| \leq \rho$; hence G is analytic near (ρ, τ) and continuity gives $G(\rho, \tau) = \tau$. By Pringsheim's theorem ρ is a singularity of A . If $G_y(\rho, \tau) \neq 1$, the analytic implicit-function theorem applied to F would continue A through ρ , a contradiction. Thus

$$G(\rho, \tau) = \tau, \quad G_y(\rho, \tau) = 1, \quad \rho \Phi_{d-1}(\rho) = 1, \quad (68)$$

and the derivative identity gives

$$G_{yy}(\rho, \tau) = \rho \Phi_{d-2}(\rho) > 0, \quad (69)$$

since the forest series includes its constant term (also for $d = 2$, where $\Phi_0 = 1$). This is the first half of Part I's (16); (67) is a second route to the finiteness of τ there.

14.2 A positive lower bound for G_z at the critical point

Here z -differentiation includes all inner functions $A(z^k)$ while y is held independent. Differentiating (56)–(57) gives, at the critical point,

$$G_z(\rho, \tau) = \frac{\tau}{\rho} + \sum_{k=2}^d (-1)^{k-1} \rho^k A'(\rho^k) \Phi_{d-k}(\rho); \quad (70)$$

the factor k from differentiating $A(z^k)$ cancels the divisor k of the logarithmic SET expression (compare Part I's (14)). Paths give $\tau/\rho \geq 1/(1-\rho)$. At the critical point $0 \leq \Phi_{d-k}(\rho) \leq \Phi_{d-1}(\rho) = \rho^{-1}$ for $2 \leq k \leq d$. Catalan domination, differentiated inside its disk, gives $A'(u) \leq T'(u) = 1/\sqrt{1-4u}$ for $0 \leq u < 1/4$, and (66) ensures $\rho^2 < 1/4$. Drop the nonnegative odd- k terms of (70), bound the even- k derivatives by the common value at ρ^2 , and enlarge their finite sum to an infinite geometric series:

$$G_z(\rho, \tau) \geq \frac{1}{1-\rho} - \frac{\rho}{(1-\rho^2)\sqrt{1-4\rho^2}} > 0. \quad (71)$$

The final strict inequality holds throughout $[0, 9/20]$: after multiplication by positive factors and squaring nonnegative quantities it is equivalent to

$$p(x) := (1+x)^2(1-4x^2) - x^2 = 1 + 2x - 4x^2 - 8x^3 - 4x^4 > 0,$$

and $p''(x) = -8 - 48x - 48x^2 < 0$, so p lies above the chord through $p(0) = 1$ and $p(9/20) = 7879/40000 > 0$. This proves transversality for every fixed $d \geq 2$ without a numerical critical-point computation.

Comparison with Part I. Proposition 2.2 is stronger: it bounds F_z below by $(I(z)/z)\eta$ along the whole branch $0 < z \leq \rho$, with the d -independent margin $\eta = 1 - 810/(319\sqrt{19})$, using $E_{d-r} \leq E_d = y/z$ where (71) uses $\Phi_{d-k} \leq \rho^{-1}$ and $\tau/\rho \geq 1/(1-\rho)$. The bound (71) holds at the critical point only. It is a second route to the transversality $F_z(\rho, \tau) > 0$ of Theorem 3.1.

14.3 The first real branch

Set $B(x) = G_y(x, A(x)) = x\Phi_{d-1}(x)$ (Part I's H). It has nonnegative coefficients, a positive coefficient of x , and $B(\rho) = 1$; therefore

$$B(x) < 1 \quad (0 \leq x < \rho). \quad (72)$$

There is no earlier characteristic point along the actual analytic branch issued from $A(0) = 0$. One can alternatively rule out a first interior crossing using $(1 - G_y)A' = G_z$ and a positive bound like (71). The argument identifies the singularity of the enumerative series before invoking a local fold; it does not select a radius by solving an unrelated algebraic system.

14.4 The square root and unique dominance

Since $F_z(\rho, \tau) = G_z(\rho, \tau) > 0$, the implicit-function theorem solves $F(z, y) = 0$ locally as $z = Z(y)$, with $Z(\tau) = \rho$, $Z'(\tau) = 0$ and $Z''(\tau) = -G_{yy}(\rho, \tau)/G_z(\rho, \tau) < 0$. Consequently $1 - Z(y)/\rho = \frac{G_{yy}}{2\rho G_z}(y - \tau)^2 Q(y)$ with analytic Q , $Q(\tau) = 1$. With an analytic square root of Q near τ , the function $(y - \tau)\sqrt{G_{yy}Q(y)/(2\rho G_z)}$ has nonzero derivative at τ and hence an analytic local inverse; this proves a convergent expansion in $t = \sqrt{1 - z/\rho}$. For real $x < \rho$ positivity gives $A(x) < \tau$, selecting

$$\beta_1 = -\sqrt{\frac{2\rho G_z}{G_{yy}}} < 0. \quad (73)$$

All derivatives in local formulas are evaluated at (ρ, τ) . (Part I reaches (17) by analytic preparation of G instead of the inverse function Z .)

To exclude other singularities on the circle, the manuscript argues exactly as the proof of Theorem 3.1: A converges uniformly on $|z| \leq \rho$, so its functional equation holds at every boundary point; $B(z)$ has nonnegative coefficients, $B(\rho) = 1$, and coefficients exactly one at z and z^2 , so equality in $|B(z)| \leq B(\rho)$ at $z = \rho e^{i\theta}$ forces $e^{i\theta} = e^{2i\theta}$, that is $e^{i\theta} = 1$. Therefore

$$|G_y(z, A(z))| = |B(z)| < 1 \quad \text{if } |z| = \rho \text{ and } z \neq \rho \quad (74)$$

(Part I's (20)), and the implicit-function theorem continues the branch through every such point. For the common domain: around ρ the selected square root is analytic in a small disk slit to the right of ρ ; the rest of the circle is compact, so finitely many implicit-function neighbourhoods cover it, and

after shrinking them (so that branches agree on overlaps) and the radial enlargement, the union with the open disk and the slit neighbourhood contains a domain (51) for any sufficiently small fixed $\phi \in (0, \pi/2)$ and some $\eta > 0$. Further shrinking keeps $|z| < \sqrt{\rho}$ wherever the inner functions are used. This proves unique dominance and the dented continuation needed for singularity analysis.

15 All-orders local and coefficient recurrences

15.1 A finite recurrence for every Puiseux coefficient

Write

$$F(\rho(1-t^2), \tau+h) = \sum_{p,q \geq 0} f_{p,q} t^p h^q. \quad (75)$$

Only even p occur. The characteristic identities give $f_{0,0} = f_{0,1} = 0$, and $f_{2,0} = -\rho G_z$, $f_{0,2} = G_{yy}/2 > 0$. Set $h = \sum_{m \geq 1} \beta_m t^m$; the coefficient of t^2 gives (73). For $m \geq 2$ substitute only the known part $h_{<m} = \sum_{i=1}^{m-1} \beta_i t^i$ and define

$$R_{m+1} = [t^{m+1}] F(\rho(1-t^2), \tau+h_{<m}), \quad \beta_m = -\frac{R_{m+1}}{G_{yy}\beta_1}. \quad (76)$$

The missing β_m first enters through $f_{0,2}h^2$, with coefficient $2f_{0,2}\beta_1\beta_m$; every other potential contribution occurs at a higher power, because $f_{0,1} = 0$ and no odd powers of t occur in (75). The denominator never vanishes. This is Part I's triangular recurrence (26), whose bracketed series is $-F$ in Part II's notation. For illustration,

$$\beta_2 = \frac{\rho G_{zy}}{G_{yy}} - \frac{G_{yyy}\beta_1^2}{6G_{yy}}, \quad (77)$$

$$\beta_3 = -\frac{1}{G_{yy}\beta_1} \left(\frac{G_{yy}\beta_2^2}{2} - \rho G_{zy}\beta_2 + \frac{\rho^2 G_{zz}}{2} + \frac{G_{yyy}\beta_1^2\beta_2}{2} - \frac{\rho G_{zyy}\beta_1^2}{2} + \frac{G_{yyyy}\beta_1^4}{24} \right). \quad (78)$$

They come from the coefficients of t^2, t^3, t^4 after substituting $z = \rho(1-t^2)$ and $h = \beta_1 t + \beta_2 t^2 + \beta_3 t^3 + \dots$ into $F(z, \tau+h) = 0$, which are respectively

$$\begin{aligned} & \frac{G_{yy}\beta_1^2}{2} - \rho G_z, & G_{yy}\beta_1\beta_2 - \rho G_{zy}\beta_1 + \frac{G_{yyy}\beta_1^3}{6}, \\ & G_{yy}\beta_1\beta_3 + \frac{G_{yy}\beta_2^2}{2} - \rho G_{zy}\beta_2 + \frac{\rho^2 G_{zz}}{2} + \frac{G_{yyy}\beta_1^2\beta_2}{2} - \frac{\rho G_{zyy}\beta_1^2}{2} + \frac{G_{yyyy}\beta_1^4}{24}. \end{aligned}$$

The coefficient of t^{m+1} is the first that determines β_m once β_1 is selected, so one must expand one degree beyond the last desired Puiseux coefficient. The recurrence applied for $m = 4, 5$ specifies β_5 without a large closed formula. For any finite order the needed derivatives of the inner functions converge absolutely because $\rho^k < \rho$ for $k \geq 2$.

15.2 Transfer coefficients at all orders

For a nonintegral exponent γ ,

$$[z^n](1-z)^\gamma = \frac{\Gamma(n-\gamma)}{\Gamma(-\gamma)\Gamma(n+1)} \sim \frac{n^{-\gamma-1}}{\Gamma(-\gamma)} \sum_{m \geq 0} H_m(\gamma) n^{-m}, \quad H_0(\gamma) = 1, \quad (79)$$

by the Gamma-ratio expansion [2, Chapter VI]. A convenient purely rational recurrence for the needed half-integer cases is

$$H_m(\gamma) = \frac{1}{m} \sum_{j=0}^{m-1} H_j(\gamma) \left\{ \binom{-\gamma-1-j}{m+1-j} - (-1)^{m+1-j} (1+\gamma) \right\}, \quad m \geq 1, \quad (80)$$

with generalized binomial coefficients given by their finite falling-product formula. The manuscript derives it by inserting $n^{-\gamma-1} \sum H_j n^{-j}$ into the exact adjacent-coefficient ratio $(n-\gamma)/(n+1)$ and equating the coefficients of $n^{-\gamma-1-(m+1)}$ (in its words, “at order n^{-m-1} ” after cancellation), where H_m has coefficient $-m$. For example $H_1(1/2) = 3/8$, $H_2(1/2) = 25/128$, $H_3(1/2) = 105/1024$, $H_1(3/2) = 15/8$.

Proposition 15.1 ([write] The two Gamma-ratio routes agree). *For every complex $\gamma \notin \{0, 1, 2, \dots\}$ and every $m \geq 0$, the numbers $H_m(\gamma)$ defined by $H_0 = 1$ and (80) are the coefficients of the asymptotic expansion $\Gamma(n-\gamma)/\Gamma(n+1) \sim n^{-\gamma-1} \sum_{m \geq 0} H_m(\gamma) n^{-m}$, and $H_m(\gamma) = \gamma_m(\gamma)$, the coefficients of Part I defined by (28)–(29).*

Proof. Write $g(n) = \Gamma(n-\gamma)/\Gamma(n+1)$. Part I’s (30) (the log-Gamma expansion) gives an asymptotic expansion $g(n) \sim n^{-\gamma-1} \sum_{k \geq 0} \gamma_k(\gamma) n^{-k}$, $\gamma_0 = 1$, to every fixed order; asymptotic expansions in the powers $n^{-\gamma-1-k}$ are unique, so it remains to show that the γ_k satisfy (80), which determines H_m from $H_0, \dots, H_{m-1}$. From $\Gamma(x+1) = x\Gamma(x)$, $g(n+1) = g(n)(n-\gamma)/(n+1)$, and $(n-\gamma)/(n+1) = 1 - (1+\gamma) \sum_{i \geq 1} (-1)^{i-1} n^{-i}$ for $n > 1$. Expand both sides to order $n^{-\gamma-1-K}$ for a fixed K : on the left, $(n+1)^{-\gamma-1-j} = n^{-\gamma-1-j} \sum_{i \geq 0} \binom{-\gamma-1-j}{i} n^{-i}$, so the coefficient of $n^{-\gamma-1-k}$ is $\sum_{j \leq k} \gamma_j \binom{-\gamma-1-j}{k-j}$; on the right it is $\gamma_k - (1+\gamma) \sum_{j < k} (-1)^{k-j-1} \gamma_j$. The remainders are $O(n^{-\gamma-2-K})$ on both sides, so the coefficients agree for $k \leq K$. The term $j = k$ on the left cancels γ_k on the right, leaving $\sum_{j < k} \gamma_j \{ \binom{-\gamma-1-j}{k-j} + (-1)^{k-j-1} (1+\gamma) \} = 0$. The summand $j = k-1$ has the factor $(-\gamma-k) + (1+\gamma) = -(k-1)$. Putting $m = k-1 \geq 1$ and solving for γ_m gives exactly (80). $\square$

Integer powers in (52) are locally analytic; any finite sum of them is a polynomial in z and contributes nothing to sufficiently large coefficients. Expand the convergent Puiseux series until its remainder is of order $(1-z/\rho)^{J+3/2}$ after subtracting the analytic terms; the transfer remainder bound then gives a relative $O(n^{-J-1})$. Combining the odd half-integer terms proves (53) with

$$C = \frac{\beta_1}{\Gamma(-1/2)} = -\frac{\beta_1}{2\sqrt{\pi}} = \sqrt{\frac{\rho G_z}{2\pi G_{yy}}} > 0 \quad (81)$$

and, for every $J \geq 0$,

$$c_J = \sum_{\ell=0}^J \frac{\beta_{2\ell+1}}{\beta_1} (-1)^\ell (3/2)_\ell H_{J-\ell}(\ell+1/2), \quad c_0 = 1, \quad (82)$$

where $(a)_\ell = a(a+1) \cdots (a+\ell-1)$, $(a)_0 = 1$, and $\Gamma(-1/2)/\Gamma(-\ell-1/2) = (-1)^\ell (3/2)_\ell$. In particular

$$c_1 = \frac{3}{8} - \frac{3}{2} \frac{\beta_3}{\beta_1}, \quad (83)$$

$$c_2 = \frac{25}{128} - \frac{45}{16} \frac{\beta_3}{\beta_1} + \frac{15}{4} \frac{\beta_5}{\beta_1}. \quad (84)$$

With Proposition 15.1, (82) is Part I’s (31) term by term, and (83)–(84) are (32)–(33). Equations (75), (76), (80) and (82) are finite all-orders algorithms, not fits to coefficient data; they complete the proof of Theorem 12.1.

15.3 Ternary and quaternary specializations

Write $B_k(z) = A(z^k)$ for the series of the given d . Expanding the signed set polynomials gives

$$\frac{G_3(z, y)}{z} = 1 + y + \frac{y^2}{2} + \frac{y^3}{6} - \frac{B_2(1 + y)}{2} + \frac{B_3}{3}, \quad (85)$$

$$\frac{G_4(z, y)}{z} = 1 + y + \frac{y^2}{2} + \frac{y^3}{6} + \frac{y^4}{24} - B_2\left(\frac{1}{2} + \frac{y}{2} + \frac{y^2}{4}\right) + \frac{B_3(1 + y)}{3} + \frac{B_2^2}{8} - \frac{B_4}{4}, \quad (86)$$

which are Part I's (5) and (6) rearranged in powers of y . Putting $y = A(z)$ reproduces the equations of the two OEIS records. The initial counts are

n	1	2	3	4	5	6	7	8	9	10	11
$d = 3$	1	1	1	2	3	6	12	25	52	113	245
$d = 4$	1	1	1	2	3	6	12	25	52	113	247

in agreement with A116379 and A116380 [7, 8] and with Part I's data.

15.4 Minimal all-orders recipe

For fixed d and target relative order J :

1. define A by the distinct-child recurrence and obtain its critical point $G = \tau$, $G_y = 1$ on the actual first branch;
2. compute the finitely many inner jets $T_{k,j}$, $2 \leq k \leq d$, $0 \leq j \leq J + 1$ (87), with tail bounds if numerical enclosures are desired;
3. build $F(\rho(1 - t^2), \tau + h)$ through total degree $2J + 2$ with the signed Newton identities;
4. choose the negative β_1 and apply (76) through β_{2J+1} ;
5. compute H_m by (80) and $c_1, \dots, c_J$ by (82);
6. take the formal logarithm (96), compute any compatible inverse order from (101), and retain the ceiling brackets (103).

The first step is justified globally by Sections 13–14; the later formal manipulations do not replace that global argument.

16 Rational interval certification of the constants

This section specifies how a finite exact computation encloses the constants of Section 15.3, and prints the enclosures. The qualitative theorem is independent of the computation. Conversely, the computation uses that theorem to identify the actual singular endpoint; it does not independently prove complex analytic continuation. The verifier (`code/02-certified-certify.py`) uses only Python's standard-library integer arithmetic, exact rational arithmetic and integer square roots. It is a computer-assisted certificate, not a formal proof-assistant development.

16.1 Independent exact enumeration and inner-series tails

The first coefficient computation builds the product (54) in increasing subtree size. Before the factor for size n is added, the coefficient at forest size $n - 1$ determines $a_d(n)$. In $(1 + uz^n)^{a_d(n)}$ the coefficient for choosing j distinct trees of size n is $\binom{a_d(n)}{j}$, and updating forest cardinalities in descending order prevents reuse of a newly updated row. A separate computation by the signed Newton recurrence (60) cross-checks every coefficient through $N = 400$. Exact divisibility, equality of the two enumerations, positivity, monotonicity and the Catalan bounds are checked.

Let a positive interval R contain ρ , let r be its upper endpoint, and take $k \geq 2$. The inner jets needed in (75) are

$$T_{k,j} = \sum_{n \geq 1} a_d(n) \rho^{kn} \binom{kn}{j}, \quad A((\rho(1 - t^2))^k) = \sum_{j \geq 0} (-1)^j T_{k,j} t^{2j}. \quad (87)$$

Put $q = 4r^k$ and $\eta_j = q((N + 2)/(N + 1))^j$. If $\eta_j < 1$, Catalan domination (61) and $\binom{kn}{j} \leq (kn)^j/j!$ yield

$$0 \leq \sum_{n > N} a_d(n) \rho^{kn} \binom{kn}{j} \leq \frac{k^j (N + 1)^j q^{N+1}}{4j! (1 - \eta_j)} : \quad (88)$$

the majorant term is $(k^j/(4j!))n^j q^n$, whose consecutive-term ratio for $n \geq N + 1$ is at most η_j . Each finite sum is evaluated by interval Horner arithmetic and enlarged by this nonnegative tail enclosure. To certify through β_5 only $j \leq 3$ and $2 \leq k \leq d$ are needed. In the reference certificate every tail bound is below $5.5 \cdot 10^{-70}$ ($d = 3$) and $8.8 \cdot 10^{-73}$ ($d = 4$).

16.2 Identification of the numerical critical root

For $d = 3, 4$ a global monotonicity argument gives a unique scalar critical root on the whole interval $0 < z \leq 9/20$. Put $u = z^2 A'(z^2)$ and $v = z^4 A'(z^4)$. From (62) and its derivative ($A(w) \leq T(w)$ and $wA'(w) \leq wT'(w) = w/\sqrt{1 - 4w}$, increasing in w), the following strict inequalities hold on that interval, certified by rational squaring at $z = 9/20$:

$$B_2 < \frac{283}{1000}, \quad u < \frac{465}{1000}, \quad B_4 < \frac{44}{1000}, \quad v < \frac{45}{1000} \quad (89)$$

(the values of the majorants at $z = 9/20$ are 0.28206, 0.46457, 0.04284 and 0.04485 to five places; the third was first printed as 0.04285 [corrected at the independent check]). For $y \geq 0$, (85)–(86) give

$$G_{3,yy} = z(1 + y) > 0, \quad G_{4,yy} = z\left(1 + y + \frac{y^2 - B_2}{2}\right) > 0.$$

Holding y fixed and dropping only nonnegative terms yields

$$G_{3,z} \geq (1 - B_2/2 - u)(1 + y) + y^2/2 + y^3/6 > 0, \quad (90)$$

$$G_{4,z} \geq (1 - B_2/2 - B_4/4 - u - v) + (1 - B_2/2 - u)y + (1/2 - B_2/4 - u/2)y^2 + y^3/6 + y^4/24 > 0. \quad (91)$$

In the second inequality the constant, linear and quadratic coefficients are strictly larger than $27/80$, $787/2000$ and $787/4000$. The retained terms omitted from these lower bounds are nonnegative because the B_k and their derivatives are nonnegative.

Moreover $G_{3,y}(z, 0) = z(1 - B_2/2) < 1$, and

$$G_{4,y}(z, 0) = z(1 - B_2/2 + B_3/3) \leq \frac{9}{20} \left(1 + \frac{283}{3000}\right) < 1,$$

where $B_3 \leq B_2$. Since $G_y(z, y)$ strictly increases to infinity with y , there is a unique positive $y_*(z)$ with $G_y(z, y_*(z)) = 1$. The function

$$M(z) = F(z, y_*(z)) \tag{92}$$

satisfies $M'(z) = G_z(z, y_*(z)) > 0$, because $F_y = 0$ along it. Theorem 12.1 supplies $M(\rho) = 0$ and $\tau = y_*(\rho)$, so the unique zero of M is the actual radius.

At each rational test endpoint, the verifier brackets y_* between 0 and 2 by strict interval signs of $G_y - 1$ and refines by bisection; an ambiguous interval sign is never used to discard a half-interval. Evaluating F over the resulting y interval encloses M . Opposite strict signs at two rational z endpoints therefore enclose ρ without a spurious-branch assumption. A fresh bisection over the whole resulting ρ interval encloses τ ; monotonicity of $y_*(z)$ is not assumed.

Remark 16.1 ([write] Two details of this argument). (i) The manuscript’s reason for $B_3 \leq B_2$, “coefficient nonnegativity and $z \leq 1$ ”, is complete: termwise $0 \leq a_d(n)z^{3n} \leq a_d(n)z^{2n}$ for $0 \leq z \leq 1$, and both series converge because $z^2 \leq 81/400 < 1/4 \leq \rho$. (With $B_3 \leq B_2$, already $1 - B_2/2 + B_3/3 \leq 1 - B_2/6 \leq 1$ gives $G_{4,y}(z, 0) < 1$, without the numerical bound $283/1000$.) [Rewritten at the independent check; see the note after this remark.] (ii) The identification uses three facts besides the bounds: $\rho \leq 9/20$ (Lemma 2.1 or (66)), so that ρ lies in the interval where M is increasing; $G_y(\rho, \tau) = 1$ with $\tau > 0$, so that $\tau = y_*(\rho)$ by uniqueness of the positive root; and y_* is differentiable, by the implicit-function theorem, since $G_{yy} > 0$ for $y \geq 0$. With these, $M(z_0) < 0 < M(z_1)$ at the two rational endpoints $z_0 < z_1$ in $(0, 9/20]$ forces $z_0 < \rho < z_1$. In the reference certificate $z_1 - z_0 = 10^{-45}$, with $M(z_0) \approx -9.0 \cdot 10^{-46}$, $M(z_1) \approx 1.3 \cdot 10^{-45}$ for $d = 3$ and $M(z_0) \approx -2.0 \cdot 10^{-46}$, $M(z_1) \approx 1.9 \cdot 10^{-45}$ for $d = 4$, each enclosed with strict sign.

(Dated note, 5 October 2026: part (i) of Remark 16.1 first read “The manuscript justifies $B_3 \leq B_2$ ‘from coefficient nonnegativity and $z \leq 1$ ’. The reason is monotonicity: A has nonnegative coefficients, so it is increasing on $[0, \rho)$, and $0 \leq z^3 \leq z^2 < \rho$. (Already $1 - B_2/2 + B_3/3 \leq 1 - B_2/6 \leq 1$ suffices for $G_{4,y}(z, 0) < 1$.” That wording presented the manuscript’s reason as deficient, and the write’s criticism was mistaken: nonnegative coefficients and $z \leq 1$ give $B_3 \leq B_2$ term by term, which is a complete proof, and monotonicity of A is the same argument restated, not a correction of it. The parenthetical, too, uses $B_3 \leq B_2$; what it avoids is only the numerical bound $283/1000$. Part (ii) is unchanged. The criticism was withdrawn at the independent check recorded at the end of Section 19.)

16.3 Outward arithmetic and elementary constants

All interval endpoints have the form L/S and H/S with integers L, H and $S = 10^{90}$. Addition and negation are exact. For products take the smallest and largest of the four integer products and divide by S with downward and upward rounding. For an interval excluding zero, reciprocal endpoints come from outward rounding of S^2/H and S^2/L . Exact rational input is rounded outward. These formulas inductively preserve enclosure, regardless of dependencies between occurrences of the same quantity. For a nonnegative interval the square-root endpoints follow from integer square roots of LS and HS , raising the upper one by one unless its square is exact. The value of π is enclosed by

Machin’s identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, using finite rational alternating sums and their next-term errors (80 and 25 terms). For logarithms set $w = (x - 1)/(x + 1)$ and use

$$\log x = 2 \sum_{j=0}^{m-1} \frac{w^{2j+1}}{2j+1} + \mathcal{R}_m, \quad |\mathcal{R}_m| \leq \frac{2r^{2m+1}}{(2m+1)(1-r^2)}, \quad (93)$$

where $r \geq |w|$ and $r < 1$; the verifier uses $m = 180$ and checks the last condition. No floating-point approximation to π , square roots or logarithms supplies mathematical evidence.

16.4 Propagation to corrections and higher orders

Substitute the interval inner jets (87) into the signed Newton polynomials and build (75) through total degree six. At the enclosed actual critical point the identities $f_{0,0} = f_{0,1} = 0$ hold exactly; their raw interval evaluations are checked to contain zero (width about 10^{-43}), and the known identities are then used as exact cancellations. This does not assert that they vanish on the whole enclosing rectangle. Compute β_1 from its negative square root and then $\beta_2, \dots, \beta_5$ by (76); each division checks a denominator bounded away from zero. Substitution residual intervals through degree six must contain zero; this is a consistency check accompanying the enclosure induction, not its replacement. Finally (81), (83), (84) and (93) enclose $C, c_1, c_2, \lambda = \log(1/\rho)$ and the inverse constant b of (95). The supplementary tests (`code/02-certified-test_certifier.py`) also check the interval primitives on 9009 random exact cases, the values of H_m above, and β_2, β_3 against the closed forms (77)–(78).

For arbitrary finite relative order J , expand the bivariate jet through total degree $2J + 2$, use inner jets $j \leq J + 1$, and compute through β_{2J+1} ; increase N to control (88) and the arithmetic and root precision as necessary. For every fixed order the tail converges geometrically because $4\rho^2 < 1$, and the recursive characteristic denominator remains nonzero. This specifies the all-orders extension; the supplied executable is specialized to degree six and $d = 3, 4$.

16.5 The certified enclosures

The following are closed rational enclosures with exact terminating-decimal endpoints, obtained by the method above and certified by the included exact integer-interval verifier. They are not rounded equalities. Here $\lambda = \log(1/\rho)$ and $b = -(3/2) \log \lambda - \log C$ are the constants of Section 17. Tables 4 and 5 are the manuscript’s.

Quantity	Lower endpoint	Upper endpoint
ρ	0.400774500477539473704620732219	0.400774500477539473704620732220
τ	1.045166273395837405445129471289	1.045166273395837405445129471290
C	0.414316870666512014232537980551	0.414316870666512014232537980552
c_1	−0.708396139596070031595088414678	−0.708396139596070031595088414677
c_2	−0.572628680061486211786162046680	−0.572628680061486211786162046679
λ	0.914356352798444759761535752942	0.914356352798444759761535752943
b	1.015426561034829120548538743296	1.015426561034829120548538743297

Table 4: A116379, $d = 3$: certified enclosures rounded outward to 30 decimals.

The full radius brackets in the reference JSON have width exactly 10^{-45} . The full enclosures for ρ, τ, C, c_1, c_2 have widths less than 10^{-35} ; Tables 4–5 round them outward to 30 places. Ninety-place

Quantity	Lower endpoint	Upper endpoint
ρ	0.397585196670872770885924531229	0.397585196670872770885924531230
τ	1.006157697606778380774781545294	1.006157697606778380774781545295
C	0.371658807505686761191129677634	0.371658807505686761191129677635
c_1	-0.453862309357400171863826908421	-0.453862309357400171863826908420
c_2	-0.562216000439483655162154638924	-0.562216000439483655162154638923
λ	0.922346036605932168162266720250	0.922346036605932168162266720251
b	1.111031252181146118715415629664	1.111031252181146118715415629665

 Table 5: A116380, $d = 4$: certified enclosures rounded outward to 30 decimals.

endpoint strings in the JSON record exact rationals and do not assert ninety accurate digits. No coefficient-asymptotic remainder bound or finite-threshold guarantee follows from these constant enclosures alone.

[write] **Added 5 October 2026, batch 103.** Table 6 prints every enclosure of `data/02-certified-certificate.json` in full, including the Puiseux coefficients $\beta_1, \dots, \beta_5$ (keys `a1`–`a5` there) and G_z, G_{yy} at the critical point, which the manuscript’s tables omit. Each entry is written as a common digit string followed by the last digits of the lower endpoint (subscript) and of the upper endpoint (superscript), both rounded outward two digits after the first place where the exact endpoints differ; the true value lies between them. For a negative quantity the subscript belongs to the more negative endpoint: for example c_1 for $d = 3$ lies between $-0.70839 \dots 727599698$ and $-0.70839 \dots 727599635$. Widths are those of the exact endpoints. Spaces group the decimals in fives.

Remark 16.2 ([write] Part I’s tables are correctly rounded). By exact rational comparison with the enclosures of Table 6: every one of the ten entries of Part I’s Table 1 lies inside the corresponding 30-place interval of Tables 4–5, and more precisely each full enclosure lies strictly within half a unit of the thirtieth decimal of the printed entry, so every printed digit string is the correctly rounded value ($D_j = c_j$, $b_j = \beta_j$). This includes the two entries discussed in the first note of Section 5 (the true C for $d = 3$ is $\dots 980551907 \dots$, the true ρ for $d = 4$ is $\dots 531229552 \dots$). Interval evaluation of (40) from the enclosures of λ, c_1, c_2 gives

$$\begin{aligned}
 d = 3 : \quad E_1 &\in 0.77474\,84199\,43689\,76129\,50794\,35644\,96402\,24129\,58_{01}^{70}, \\
 E_2 &\in 2.17165\,20579\,71796\,11833\,55612\,75953\,08307\,6073_{946}^{963}, \\
 d = 4 : \quad E_1 &\in 0.49207\,37893\,85523\,88787\,58880\,80793\,10675\,35119\,88_{50}^{78}, \\
 E_2 &\in 1.52147\,03882\,82210\,65137\,47100\,98590\,36465\,6153_{815}^{822},
 \end{aligned}$$

and Part I’s 20- and 19-decimal values of E_1, E_2 (Section 6) are their correct roundings. These are computations from the certificate, not new proofs; they inherit its trust boundary (computer-assisted, and the use of Theorem 3.1 to identify ρ).

17 Asymptotic inversion with the rounding retained

Define the integer threshold

$$N_d(T) = \min\{n \geq 1 : a_d(n) \geq T\}, \quad T > 0. \quad (94)$$

Adding a unary root is an injection, so $a_d(n+1) \geq a_d(n)$ for all n . Theorem 12.1 gives $a_d(n+1)/a_d(n) \rightarrow \rho^{-1} > 1$, so the sequence is eventually strictly increasing and unbounded, and the

	enclosure	width
A116379, $d = 3$		
ρ	0.40077 45004 77539 47370 46207 32219 34740 07631 8924 ⁸⁰ ₇₀	$1.0 \cdot 10^{-45}$
τ	1.04516 62733 95837 40544 51294 71289 20906 23353 965 ⁸⁶ ₄₁	$4.5 \cdot 10^{-44}$
$G_z(\rho, \tau)$	2.20583 85694 69025 03803 18421 85039 16193 73722 69 ⁷⁴ ₅₈	$1.5 \cdot 10^{-43}$
$G_{yy}(\rho, \tau)$	0.81965 04916 13727 66408 44577 13351 64056 00647 978 ⁵⁶ ₃₅	$2.0 \cdot 10^{-44}$
C	0.41431 68706 66512 01423 25379 80551 90734 83055 502 ³³ ₆₃	$1.8 \cdot 10^{-44}$
c_1	-0.70839 61395 96070 03159 50884 14677 28095 72759 96 ³⁵ ₉₈	$6.2 \cdot 10^{-43}$
c_2	-0.57262 86800 61486 21178 61620 46679 51243 22372 ⁷⁷ ₉₁	$1.3 \cdot 10^{-41}$
λ	0.91435 63527 98444 75976 15357 52942 35245 90374 1733 ⁶⁰ ₃₄	$2.5 \cdot 10^{-45}$
b	1.01542 65610 34829 12054 85387 43296 59301 56192 4 ⁹¹ ₈₉	$4.8 \cdot 10^{-44}$
β_1	-1.46871 50658 15963 70493 25307 17216 96234 97520 83 ⁷⁴ ₈₂	$6.5 \cdot 10^{-44}$
β_2	0.93052 47133 16070 52313 06735 09292 04842 31876 25 ⁶¹ ₄₂	$1.9 \cdot 10^{-43}$
β_3	-1.06080 01549 81068 66544 87619 20369 88431 68940 ⁷⁹ ₈₁	$5.6 \cdot 10^{-43}$
β_4	0.86232 42707 61499 26723 17876 90588 15402 40230 6 ⁷⁵ ₅₈	$1.6 \cdot 10^{-42}$
β_5	-0.49483 03080 17989 61748 83456 44803 65949 37519 9 ²⁸ ₇₇	$4.8 \cdot 10^{-42}$
A116380, $d = 4$		
ρ	0.39758 51966 70872 77088 59245 31229 55260 91912 6980 ⁶⁰ ₅₀	$1.0 \cdot 10^{-45}$
τ	1.00615 76976 06778 38077 47815 45294 75819 09840 577 ⁵³ ₄₁	$1.1 \cdot 10^{-44}$
$G_z(\rho, \tau)$	2.09856 66204 36446 98331 91531 98306 88101 41966 6 ⁸¹ ₇₉	$5.7 \cdot 10^{-44}$
$G_{yy}(\rho, \tau)$	0.96135 59976 29576 01020 70836 97169 69343 86803 987 ⁴⁶ ₃₃	$1.2 \cdot 10^{-44}$
C	0.37165 88075 05686 76119 11296 77634 94289 87038 536 ⁶⁶ ₅₈	$6.8 \cdot 10^{-45}$
c_1	-0.45386 23093 57400 17186 38269 08420 63109 56911 57 ⁶⁶ ₉₃	$2.5 \cdot 10^{-43}$
c_2	-0.56221 60004 39483 65516 21546 38923 69892 44636 3 ⁰⁶ ₆₂	$5.5 \cdot 10^{-42}$
λ	0.92234 60366 05932 16816 22667 20250 92941 67672 3254 ⁹⁰ ₆₄	$2.5 \cdot 10^{-45}$
b	1.11103 12521 81146 11871 54156 29664 11488 13787 94 ²² ₁₉	$2.2 \cdot 10^{-44}$
β_1	-1.31749 61691 72812 80730 03444 20338 28942 05344 353 ⁰¹ ₂₇	$2.4 \cdot 10^{-44}$
β_2	0.64123 77359 09655 16110 66432 56599 27453 93667 73 ¹² ₀₄	$6.9 \cdot 10^{-44}$
β_3	-0.72801 52782 33403 73385 06882 67526 37995 77355 76 ⁰¹ ₂₃	$2.1 \cdot 10^{-43}$
β_4	0.60853 66639 94243 16927 60797 36207 72101 46119 87 ⁷³ ₁₂	$6.0 \cdot 10^{-43}$
β_5	-0.27986 72193 76854 09137 86233 54511 19191 74825 1 ²³ ₄₁	$1.8 \cdot 10^{-42}$

Table 6: [write] Every certified enclosure of the reference certificate, in full (format in the note above).

threshold is well defined (Part I's (48), with $T = y$). Put

$$\alpha = \frac{3}{2}, \quad \lambda = \log(1/\rho), \quad L = \log T, \quad \ell = \log L, \quad b = -\alpha \log \lambda - \log C, \quad (95)$$

and define d_j by the formal logarithm

$$\log\left(1 + \sum_{j \geq 1} c_j x^j\right) = \sum_{j \geq 1} d_j x^j, \quad (96)$$

so $d_1 = c_1$ and $d_2 = c_2 - c_1^2/2$. At integers $n \rightarrow \infty$,

$$\log a_d(n) = \lambda n - \alpha \log n + \log C + \sum_{j=1}^J d_j n^{-j} + O(n^{-J-1}). \quad (97)$$

This is a statement about integer coefficients. A real formal inverse is an intermediate asymptotic expression, not an extension of the combinatorial count to all real n . Seek

$$f_M(L) = \frac{1}{\lambda} \left(L + \alpha \ell + b + \sum_{m=1}^M P_m(\ell) L^{-m} \right). \quad (98)$$

With $u = \alpha \ell + b$, the first two numerator corrections are

$$P_1(\ell) = \alpha u - \lambda d_1, \quad (99)$$

$$P_2(\ell) = \alpha P_1 - \frac{\alpha}{2} u^2 + \lambda d_1 u - \lambda^2 d_2. \quad (100)$$

For all orders, set $x = 1/L$ and $s = u + \sum_{m \geq 1} P_m x^m$; the triangular recursion is

$$P_m = [x^m] \left\{ \alpha \log(1 + xs) - \sum_{j=1}^m d_j \lambda^j x^j (1 + xs)^{-j} \right\}, \quad m \geq 1. \quad (101)$$

Only P_i with $i < m$ can affect this coefficient, because s always occurs multiplied by x . To derive (101), substitute $n = (L + s)/\lambda$ into (97), use $\log n = \log L - \log \lambda + \log(1 + xs)$, and cancel the leading terms; this also shows by induction that P_m is a polynomial in ℓ of degree at most m . The first-correction formula, often the most useful, is

$$f_1(L) = \frac{1}{\lambda} \left[L + \frac{3}{2} \log L + b + \frac{\frac{9}{4} \log L + \frac{3}{2} b - \lambda c_1}{L} \right]; \quad (102)$$

in particular the sign in front of λc_1 is negative.

Theorem 17.1 (All-orders integer inverse brackets). *For every fixed $d \geq 2$ and integer $M \geq 0$ there are constants $K > 0$ and T_0 such that, for $T \geq T_0$, writing $L = \log T$ and $\epsilon_M(L) = K(\log L)^{M+1}/L^{M+1}$,*

$$\lceil f_M(L) - \epsilon_M(L) \rceil \leq N_d(T) \leq \lceil f_M(L) + \epsilon_M(L) \rceil. \quad (103)$$

For sufficiently large T the endpoints are equal or adjacent integers. In particular $N_d(T) = (L + \frac{3}{2} \log L - \frac{3}{2} \log \lambda - \log C)/\lambda + O(1)$, and the real interval inside the ceilings for (102) has width $O((\log L)^2/L^2)$.

Proof. Choose in (97) sufficiently many terms for the desired M and call the resulting smooth truncated expression $h(n)$. Its derivative satisfies $h'(x) = \lambda + O(1/x)$, so $h'(x) \geq \lambda/2$ for large x . Formal substitution and (101) give $h(f_M(L)) - L = O((\log L)^{M+1}L^{-M-1})$. The integer error in (97) is bounded by a constant times a suitable negative power of n , and is no larger than this scale once $n \asymp L$. Increasing K supplies lower and upper comparison points $f_M(L) \mp \epsilon_M(L)$ with a strict margin covering both errors. Every sufficiently large integer below the lower point has logarithmic count less than L ; every integer at or above the upper point has logarithmic count at least L ; the finitely many small integers are below threshold once T is large. Taking ceilings proves (103). Finally $2\epsilon_M(L) < 1$ eventually, so the two ceilings differ by at most one. $\square$

Remark 17.2 ([write] The same theorem as Part I's). Theorem 17.1 is Proposition 7.1 with the pure-logarithmic expansion (42), in another normalization: Part I expands in $L_I = \log(T/C)$ and $h = \log(L_I/\lambda)$, Part II in $L = \log T$ and $\ell = \log L$, and Part I's p, D_j, L_2 are α, c_j, d_2 . A second proof from Part I runs as follows. Fix M and take Part I's model order $R \geq M$, so that its $\log M_R(x) = \lambda x - \alpha \log x + \log C + \sum_{j \leq M} d_j x^{-j} + O(x^{-M-1})$ with the true $d_1, \dots, d_M$. Put $\lambda x = L + s$; then $\log M_R(x) - L = s - \alpha \log(1 + s/L) - u + \sum_{j \leq M} d_j \lambda^j L^{-j} (1 + s/L)^{-j} + O(L^{-M-1})$ for bounded s/L , and (101) is precisely the statement that $s = s_M := u + \sum_{m \leq M} P_m L^{-m}$ makes this $O((\log L)^{M+1}L^{-M-1})$ (the first neglected coefficient is a polynomial of degree at most $M+1$ in ℓ). Since $(\log M_R)' \geq \lambda/2$ for large arguments (36), the inverse $x_R(T)$ of Part I satisfies $|\lambda x_R(T) - \lambda f_M(L)| \leq K_1(\log L)^{M+1}L^{-M-1}$ for large T . Part I's bracket (49) around $x_R(T)$ has half-width $K_{d,R}x_R^{-R-1} \leq K_2L^{-M-1}$, because $x_R(T) \sim L/\lambda$ and $R \geq M$. With $K = K_1/\lambda + K_2$ and $L \geq e$ we get $f_M - \epsilon_M \leq x_R - K_{d,R}x_R^{-R-1}$ and $f_M + \epsilon_M \geq x_R + K_{d,R}x_R^{-R-1}$, and monotonicity of the ceiling turns (49) into (103). By uniqueness of expansions in the scale $(\log L)^a L^{-m}$, re-expanding Part I's (42) in L reproduces the P_m of (101); for $m = 1, 2$ the write checked this symbolically (Section 11.2), for example $P_1 = \alpha u - \lambda d_1$ is Part I's $P_1(h) = p^2 h - \lambda D_1$ after $L_I = L - \log C$, $h = \ell - \log \lambda + O(1/L)$. Both inversions are instances of the transseries volume's results, as the note at the end of Section 7 says.

Remark 17.3 (What an inverse expansion does not say). In general $N_d(T) \neq f_M(\log T) + o(1)$: an integer-valued threshold necessarily retains a rounding effect. Nor do the existential constants K, T_0 of Theorem 17.1 constitute an algorithmically usable finite- T certificate. The rational constant enclosures of Section 16 do not supply explicit coefficient-remainder constants or a numerical range for that theorem.

18 Exact finite thresholds and reproducibility

18.1 An exact threshold algorithm

For a specified positive integer T , a complete certificate for (94) needs no asymptotic remainder estimate:

1. enumerate exact coefficients through a bound M , initially for example $M = 32$, using (54) or (60);
2. if $a_d(M) < T$, double M and extend or recompute the exact list;
3. once $a_d(M) \geq T$, binary-search the monotone exact list for its first coefficient at least T ;
4. return n and the integer inequalities $a_d(n-1) < T \leq a_d(n)$, with the convention $a_d(0) = 0$.

The unary-root injection proves the monotonicity the search needs. Unboundedness from Theorem 12.1, or directly the spine subclass (64), proves termination. A formal inverse may suggest a starting bound, but the proof of the output uses only integer comparisons. For a real noninteger input, exact or certified comparisons to that input must be specified; a floating-point approximation must not silently be treated as an exact real number. (Part I’s Section 9 has the exact enumeration but no threshold search.)

18.2 Computational scope and trust boundary

The delivered package contained the mathematical source, its PDF, the unchanged integer-interval verifier, the reference certificate, exact coefficient exports through 400 and supplementary verifier tests; the repository ships the verifier, the certificate, the tests and the release tools (see the shipped-files note below). The executable certificate is restricted to $d = 3, 4$ and local degree six. General d and arbitrary orders are proved and algorithmically specified, not claimed as already implemented and exercised by that executable. The certificate replays with Python 3 and no third-party dependencies: run the program from a temporary directory, compare its standard output byte for byte with the frozen reference JSON, and repeat with optimization enabled. A build script compiles the article in a fresh temporary output directory. The guarantees have distinct roles:

- the mathematical proof establishes the model, actual radius, unique square-root singularity, every fixed asymptotic order, and the asymptotic inverse brackets;
- the numerical certificate encloses selected constants and local coefficients for the two specified sequences;
- exact coefficient comparisons certify finite thresholds when that algorithm is run for a stated input;
- hashes establish file identity; replays and tests establish the outcomes of those runs, not a proof-assistant guarantee or a universal software-correctness theorem.

No finite- n asymptotic error bound, explicit inverse remainder constant, global literature-priority conclusion, upload, submission or publication is part of these guarantees.

[write] **Added 5 October 2026, batch 103.** *Shipped files.* The repository ships, byte-identical to the delivery and with the prefix 02-certified-, five programs in `code/` (`certify.py` and `test_certifier.py`, delivered in `certificate/`; `verify_package.py`, `test_package.py` and `build.sh`) and three records in `data/` (`certificate.json` and `tests.json`, delivered in `certificate/`; `release_checks.json`). Not shipped (all in the arrival commit): the manuscript sources and PDF, the delivered README, the checksum manifest SHA256SUMS (verified at intake) and the two coefficient exports `a_d3_through_400.json` and `a_d4_through_400.json`, which `certify.py --coefficients-out` regenerates in seconds, whose SHA-256 the certificate records, and whose integers are Part I’s `data/identity-checks.json`. The prefixed names break the delivered imports (`import certify`, `import verify_package`), and `verify_package.py` needs SHA256SUMS and the delivered tree beside it; the report’s `README.md` gives a rerun route on a scratch copy with the prefix removed, and warns that on Windows the programs write CRLF, so outputs must be compared modulo carriage returns.

19 Further questions and research (Part II)

Vladimir’s standing rule of 4 October 2026: unproved claims are recorded here, with their source, sketch and what is missing. The manuscript’s own “Further research” paragraph names three

extensions as outside the proved scope, each requiring additional arguments rather than a change of parameter; they are questions 1, 3 and 4.

1. **Effective remainders and thresholds** (manuscript; Part I’s question 1, second half). Effective finite- n remainder constants and a certified starting index would turn the brackets of Theorem 17.1 (or (49)) into fast numerical threshold certificates, reducing reliance on exact enumeration. Missing: explicit K and T_0 ; this needs an explicit transfer remainder in the Δ -domain, hence explicit η, ϕ and a bound for A on its boundary, none of which Part I or Part II provides.
2. **Certified corrections of higher order and other caps** (manuscript, Section 16). The certificate encloses c_1, c_2 and $\beta_1, \dots, \beta_5$ for $d = 3, 4$ only. The extension to c_J for $J \geq 3$ and to other d is specified (jets through degree $2J + 2$, inner jets $j \leq J + 1$, larger N) but not implemented or run. For $d \geq 5$ the global identification of Section 16.2 would also need new bounds like (90)–(91), since it is proved only for $d = 3, 4$.
3. **Uniformity as d grows** (manuscript; Part I’s question 2). Estimates uniform in d would connect the fixed-outdegree expansions with the unrestricted class (A004111) and the maximum-outdegree behaviour of a large random identity tree. Neither Part proves anything uniform in d .
4. **Unrooted identity trees with a bound on total degree** (manuscript; new). Unrooted (free) identity trees with every vertex degree at most d call for a separate treatment of automorphisms and root positions. The manuscript suggests equivariant dissymmetry as a route, but proves neither an unrooted counting formula nor its asymptotics, and nothing in either Part addresses it. (Haigh–Kennedy–Quintas [6] count identity trees of bounded degree with fixed diameter; their full text, and so their degree convention, has not been checked, and comparing conventions is Part I’s question 5.)
5. **Priority** (both manuscripts). Four older works remain unchecked at full-text level: Meir–Moon–Mycielski [5] and Haigh–Kennedy–Quintas [6], cited by Part I, and Labelle [9] and Kennedy–McKeon–Palmer–Robinson [10], cited by Part II after abstract-level checks only. Part II’s manuscript does not cite Meir–Moon–Mycielski. Neither Part claims priority.
6. **The OEIS status** (manuscript). The statement that the A116379 and A116380 records “explicitly retain a question about the leading form” was inspected by the manuscript on 2 October 2026 and re-checked by the intake on 5 October 2026: both records still state that it is not known whether these trees have the asymptotic form $C\rho^{-n}n^{-3/2}$ [updated at the independent check].

No claim of the manuscript was found to be mathematically wrong. The one demonstrably wrong statement is bibliographic, the author of [6] (Remark 11.1); a packaging overstatement (the delivered README’s “any working directory” replay does not pass on Windows except modulo line endings) is disclosed in the report’s README.md.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of Part II, made by the intake after the write (a2dc5d320), reread every item of the write that carries a proof or a transcription: Proposition 15.1, Remarks 11.1, 13.1, 16.1, 16.2 and 17.2, and Table 6. It also reread the certificate method of Section 16, both as printed and as coded in code/02-certified-certify.py. It found every item valid, no counterexample, and no gap in

any proof chain. One statement of the write was wrong. Part (i) of Remark 16.1 presented the manuscript's reason for $B_3 \leq B_2$, "coefficient nonnegativity and $z \leq 1$ ", as deficient, although it is a complete termwise proof. That criticism is withdrawn: part (i) is rewritten where it stands, with a dated note after the remark keeping its first wording. While the check was applied, one five-place value added by the write in Section 16.2 was also found wrong. The plane-tree majorant of B_4 at $z = 9/20$ is $T((9/20)^4) = 0.0428416\dots$, which rounds to 0.04284, not to the printed 0.04285. It is corrected where it stands; the certified bound $B_4 < 44/1000$ is unaffected. The entry [5] now gives the volume and pages, which the Crossref record of its DOI gives as **35**(2), 142–155 (October 1983). Both OEIS records were re-checked on 5 October 2026 and still state that it is not known whether these trees have the asymptotic form $C\rho^{-n}n^{-3/2}$. The bracketed additions in the first note of Part I, after Theorem 12.1, in question 6 above and in the entries [7], [8] record this. No other statement or number changed, and no label was added or renumbered.

The check used neither the delivered programs nor the write's. Its own dynamic program for the counts builds sets of distinct trees by child size, in a loop order different from the certifier's. Its counts agree with:

- a brute-force enumeration of canonical tree forms for $n \leq 13$ and $d = 2, 3, 4$;
- the OEIS b-files of A116379 and A116380 for $n \leq 1000$;
- Part I's `data/identity-checks.json` for $n \leq 400$;
- the printed $a_2(n)$ for $n \leq 20$.

It then recomputed every certified constant ($\rho, \tau, C, c_1, c_2, \lambda, b, \beta_1, \dots, \beta_5, G_z, G_{yy}, E_1, E_2$ for $d = 3, 4$) at 100 digits, with inner sums through $n = 1000$. Its route differs from the manuscript's:

1. Newton's method on $F = 0, G_y = 1$;
2. the bivariate Taylor jet of F at (ρ, τ) to total degree 11;
3. reversion of the local inverse in t to get $\beta_1, \dots, \beta_5$;
4. the explicit transfer coefficients of [2, Theorem VI.1] in place of (80).

As internal checks, $\beta_1 + \sqrt{2\rho G_z/G_{yy}}$ is below 10^{-100} and $C = \sqrt{\rho G_z/(2\pi G_{yy})}$. The counts check c_1 and c_2 as well: the quantity $n^3(a_d(n)/(C\rho^{-n}n^{-3/2}) - 1 - c_1/n - c_2/n^2)$ settles at $n = 250, 500, 1000$, as a bounded next coefficient requires. The values are $-4.671, -4.604, -4.571$ for $d = 3$ and $-3.830, -3.771, -3.743$ for $d = 4$. Exact rational comparison of the recomputed values with every printed enclosure gave the following:

- all 28 enclosures of the certificate JSON contain them strictly, with both margins about half the width;
- each of the 28 rows of Table 6 contains both the recomputed value and the JSON interval, and its printed width is the JSON width to two significant figures;
- all 14 thirty-place intervals of Tables 4–5 contain them;
- the ten entries of Table 1 and Part I's four values of E_1, E_2 lie within half a unit of their last printed place, the closest at -0.492 and $+0.488$ units, well clear of the boundary at the enclosure widths;
- the four E enclosures of Remark 16.2 contain them.

The recorded endpoint signs of M are what the recomputed ρ predicts: $G_z \cdot (\rho - z_0) \approx 9.0 \cdot 10^{-46}$ for $d = 3$ and $2.0 \cdot 10^{-46}$ for $d = 4$.

For Proposition 15.1 the check reread the proof line by line and confirmed symbolically in γ , for $m \leq 7$, that (80) equals Part I's Bernoulli route. A third route, against Γ itself, gives $n^6(\Gamma(n - \gamma)n^{\gamma+1}/\Gamma(n + 1) - \sum_{k \leq 5} H_k n^{-k}) \rightarrow H_6$ for $\gamma = 1/2, 5/2, -7/3$ and $3/10 + 11i/10$. The proof never uses the exclusion of $\gamma \in \{0, 1, 2, \dots\}$, so the recurrence also holds for negative and complex γ . For Remark 17.2 it re-derived each step, and it checked symbolically that P_1 and P_2 agree with Part I's after re-expansion. A control with the sign of the u^2 term of P_2 flipped gives a nonzero difference, so the comparison can fail. For Remark 13.1 exact rational evaluation gives $x(1 + S(x)) - 1 \approx -1.05 \cdot 10^{-7}$ at 0.4464104 and $+7.19 \cdot 10^{-7}$ at 0.4464105, with $r = 0.44641041274508678\dots$, and reproduces the rational number (65). Crossref confirms the authors, title and data of [6] (Remark 11.1) and the data of [9] and [10].

In the certificate method the check confirmed the following:

- the outward rounding of products, reciprocals and integer square roots;
- the Machin enclosure of π ;
- the remainder bound (93), about 10^{-133} for λ ;
- the Catalan tail majorant (88) with its ratio bound, the largest tails in the JSON being $5.455 \cdot 10^{-70}$ ($d = 3$) and $8.740 \cdot 10^{-73}$ ($d = 4$);
- that the jet orders suffice for β_5 ;
- that the two critical-point identities are imposed as exact zeros only after their enclosures are checked to contain zero;
- the bounds (89), with majorants 0.2820551, 0.4645669, 0.0428417, 0.0448491 at 9/20;
- the constants 27/80, 787/2000, 787/4000 and $p(9/20) = 7879/40000$;
- (90)–(91), re-derived by differentiating (85)–(86);
- the uniqueness of y_* and of the zero of M .

This was a careful reading with independent high-precision computation, not a formal verification.

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